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Parametric assessment of a building-integrated PV/T system with a Nanofluid/NEPCM spectral splitter

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ABSTRACT

This study aims to inspect the energetic and exergetic performance of a hybrid system involving a heat exchanger and a building-integrated concentrating photovoltaic/thermal (CPV/T) system equipped with a spectral splitting unit. The used spectral unit is a combination of glycerol/silver nanofluid (NF) and dotriacontane/gold nano-enhanced phase change material (NEPCM), which are used as spectral splitter and absorptive filter, respectively. The nanofluid, which is heated due to the absorption of thermal energy from the sun's radiation, enters the heat exchanger to heat the water leaving the CPV/T system, which is preheated due to the absorption of heat from the photovoltaic panel. Mie and Rayleigh scattering principles are utilized to model the optical characteristics of the NF and NEPCM. The effect of the CPV/T system length, overall heat transfer coefficient of the heat exchanger, the mass flow rate of NF and water, as well as the thickness of NEPCM, nanofluid, coolant, and absorber layers on the average yearly energy and exergy efficiencies of the system is investigated. Among the studied cases, the hybrid system displayed its peak average annual energy and exergy efficiencies at 123.9 % and 17.08 %, respectively.

1. Introduction

Statistics provided by the International Energy Agency show a 26 % growth in solar electricity production using photovoltaic (PV) panels in 2022 compared to the previous year [1]. With this growth, the solar electricity produced has reached 270 TW h, equivalent to 4.5 % of the total electricity produced in the world [1]. Despite this and other statistics that indicate the global acceptance of PV technology and the development of its application worldwide, there are still many challenges regarding the application of PV systems. The efficiency of most commercial PV panels is about 17–20 % [2], much lower than that of fossil power plants. Still, a point that makes this problem more acute is the reduction of the efficiency of PV panels as their temperature boosts [3]. These problems caused some researchers to seek to improve PV technology by introducing new semiconductor materials, such as CdTe [4], GaAs [5], CIGS [6], and perovskites [7], and others sought to find ways to obtain excess heat from the PV panels. Researchers have put forward several practical recommendations for cooling PV panels, such as employing PCMs [8], thermoelectric generators [9], magnetic fields [10], heat pipes [11], ultrasound fields [12], jet impingement [13], water spray [14], vortex tube [15], fluid flow [16], and corrugated surfaces [17].

When the heat obtained from the solar panel is transferred to a fluid flow through a thermal collector, the corresponding system is known as a photovoltaic/thermal (PV/T) system [18]. In investigating these systems, scholars have discussed how a range of factors,

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Nomenclature

A	heat transfer surface area of heat exchanger (m^2)
BIPV/T	Building-Integrated Photovoltaic/Thermal
C	Concentration ratio
CF	Coolant Fluid
C_p	Specific heat capacity (J/kg·K)
CPV/T	Concentrating Photovoltaic/Thermal
h_c	combined convection-conduction heat transfer coefficient ($W/m^2·K$)
h_{cond}	conduction heat transfer coefficient ($W/m^2·K$)
h_{conv}	convection heat transfer coefficient ($W/m^2·K$)
h_{PCM}	Latent heat of PCM (J/kg)
h_r	radiation heat transfer coefficient ($W/m^2·K$)
I_0	Radiation intensity
k	Thermal conductivity (W/m·K)
L	Length of the CPV/T unit (m)
n	refractive index
m	the ratio of the refractive index of two adjacent layers (>1)
$\dot{m}$	Mass flow rate (kg/s)
NEPCM	Nano-Enhanced Phase Change Material
NF	Nanofluid
PCM	Phase Change Material
PV	Photovoltaic
r	particle size
SSCPV/T	Spectral Splitting Concentrating Photovoltaic/Thermal
T	Temperature (K)
T_a	Ambient temperature (K)
$T_{PCM,m}$	Phase change temperature of PCM (K)
T_{sun}	Temperature of the sun (K)
$T_{cf,he}$	Temperature of water leaving the heat exchanger (K)
$T_{cf,out}$	Temperature of water leaving the SSCPV/T unit (K)
U	Overall heat transfer coefficient ($W/m^2·K$)
W	Width (m)
w_j	quadrature weights
X	ratio of the particle size (r) to the wavelength of the incoming light (λ)

Greek letters

α	Absorptance
β	Extinction coefficient (1/m)
$\beta_{NEPCM,\lambda}$	Extinction coefficient of NEPCM at wavelength λ (1/m)
$\beta_{nf,\lambda}$	Extinction coefficient of nanofluid at wavelength λ (1/m)
λ	wavelength (nm)
δ	Thickness (m)
ε	Effectiveness of the heat exchanger
η	First-law efficiency
μ	Dynamic viscosity (Pa·s)
μ_j	direction cosines
ρ	Density (kg/m^3)
σ	Second-law efficiency
ω	single scattering Albedo
τ	Transmittance

Subscripts

abs	absorber
bp	black plate
cf	coolant fluid
el	electrical
g	glass
he	heat exchanger exit
in	inlet

l	liquidus
NEPCM	Nano-Enhanced Phase Change Material
of	optical fluid
PV	photovoltaic cell
PCM	phase change material
s	solidus
th	thermal
out	outlet
ov	overall

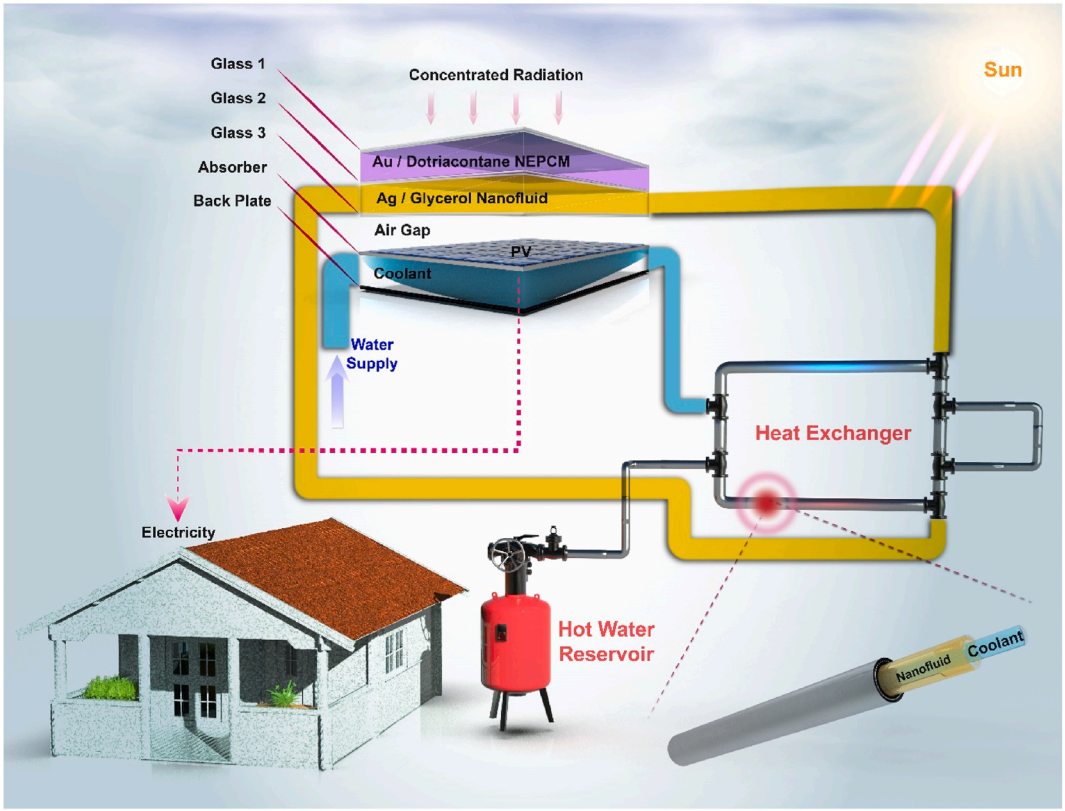
such as the cooling fluid type, collector geometry, weather conditions, operational settings, etc. influence the performance of PV/T systems [19–21]. Researchers proposed using concentrators to enhance the temperature of the heat transfer fluid leaving the PV/T systems [22]. Although this idea was effective in improving the coolant temperature, it led to a notable decline in the efficiency of the PV cells [23], the reason of which is explained below. Solar cells can convert only the sunlight with spectral wavelengths between 400 nm and 1100 nm into electricity. The other parts of the spectrum are converted into thermal energy, which causes the solar cell to heat up and reduce its electrical conversion electricity by 4–6.5 % for every 10 °C increment in operating temperature for silicon cells [24]. To solve this problem, researchers suggested using beam splitters or absorptive optical filters, whose task is to separate the sunlight spectrum into wavelengths of less and more than 1100 nm; the former and the latter are used to heat a fluid and to produce electricity in PV panels, respectively [25]. One commonly used beam splitter in CPV/T systems is liquids because they serve the dual purpose of being both a beam splitter and a heat transfer fluid simultaneously. Table 1 presents a summary of some recent contributions to the SSPV/T literature.

As absorption filters, PCMs have been widely noticed by researchers because they can absorb and store a large amount of energy in the form of latent heat [26]. A technique aimed at increasing the efficiency of CPV/T systems by using a nanofluid (NF) as a spectral splitter and a nano-enhanced PCM (NEPCM) as an absorptive filter was applied by Yazdanifard et al. [26]. They investigated different arrangements for this system and evaluated their efficiencies numerically. Their investigations revealed that placing the NF layer above the NEPCM layer results in the highest output temperature and energy and exergy efficiency. They also concluded that dispersing nanoparticles in PCM or optical fluid significantly enhances their temperature due to the substantial absorption of radiation by the nanoparticles. Additionally, placing the NEPCM at the bottom of the optical fluid results in the best performance from an energy storage perspective. These enhancements over the base fluid demonstrate the significant potential of nanofluids in improving system efficiency and performance. Meraje et al. [27] conducted an experiment to evaluate the effectiveness of a Fresnel lens-based CPV/T system including an NF spectral splitter. The overall efficiency of the system containing ZnO NF with a volume fraction of 0.00036–0.0089 % was reported in the range of 50.35–85 %. Huang et al. [28] experimentally inspected the stability, optical features, and usability of propylene glycol/Ag-SiO_2 nanofluids in CPV/T systems as a spectral splitter. It was found that subjecting the NF to elevated high-temperature conditions for 60 days had no impact on its stability and optical properties, and using it in silicon PV panels is associated with 43 % filtering efficiency. Elharoun et al. [29] measured the outcomes of a CPV/T device with a fluid spectral filter. The impact of the air gap thickness between the filter and the PV panel and the tilt angle of the concentrator on the system operation was inspected. In the optimal state, the application of the filter was associated with a 3.9 % elevation in the system's electrical efficiency. Zhang et al. [30] presented the feasibility of a CPV/T system with a spectral splitter and absorption heat pump. The use of a splitter led to an increase in the electrical efficiency of the CPV/T device and an increase in the water temperature, which was increased by the geothermal energy in the heat pump, increased more. The optimal device showed overall energy and exergy efficiencies of 102.14 % and 37.49 %, respectively.

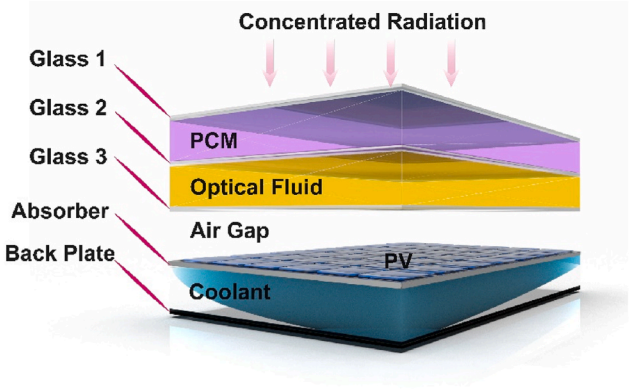
Table 1
Summary of the literature on SSPV/T systems.

Study	Configuration Details	Efficiency/Performance Results	Key Findings
Yazdanifard et al. [26]	- NF as spectral splitter and NEPCM as an absorptive filter	- Highest output temperature - Highest energy and exergy efficiency	Dispersing nanoparticles in PCM or optical fluid significantly enhances their temperature due to the substantial absorption of radiation by the nanoparticles.
Meraje et al. [27]	- Fresnel lens-based CPV/T system - ZnO NF with volume fraction 0.00036–0.0089 % as the spectral splitter	- Overall efficiency: 50.35–85 %	Effective use of ZnO nanofluid in a CPV/T system with varying volume fractions.
Huang et al. [28]	- Propylene glycol/Ag-SiO2 nanofluids as spectral splitter	43 % filtering efficiency	The stability and optical properties of nanofluid remain intact even at elevated temperatures for 60 days.
Elharoun et al. [29]	- CPV/T device with a fluid spectral filter	Electrical efficiency: +3.9 %	Optimal air gap thickness and tilt angle can enhance the electrical efficiency of the system.
Zhang et al. [30]	- CPV/T system with spectral splitter and absorption heat pump	- Energy efficiency: 102.14 % - Exergy efficiency: 37.49 %	Spectral splitter improves electrical efficiency and increases water temperature, enhanced further by geothermal energy.

By examining the results presented by Yazdanifard [26], it is found that, firstly, the temperature of the spectral splitter reaches more than 150 °C. Secondly, the researchers focused on evaluating the efficiency of the SSCPV/T system and did not consider a specific application for it. For this purpose, water is first passed under the PV panel as a coolant, and after leaving the corresponding channel, it is sent to a heat exchanger to be reheated by taking heat from the spectral splitter. Therefore, current research has been done to determine whether the proposed system (combination of the SSCPV/T and the internal heat exchanger) can provide the electricity and hot water needed by a building during the year. The system is numerically run for 24 h on the 21st day of each 12 months. The annual



(a)



(b)

Fig. 1. Schematic sketch of the examined SSCPV/T-heat exchanger system.

averaged effect of several parameters, such as the length of the SSCPV/T system, the overall heat transfer coefficient of the heat exchanger, the thickness of PCM, spectral splitter, coolant, and air gap layers, along with the mass flow rate of spectral splitter and water on the performance parameters of the system has been investigated. To analyze the obtained results, both the energy and exergy perspectives have been used. While energy analysis provides valuable insights, it often falls short in accurately reflecting performance, as it does not account for the quality of energy flows. Exergy analysis overcomes this limitation by simultaneously considering both the quantity and quality of energy flows [31], offering precise information on the thermodynamic efficiency of solar energy systems. The application of exergy analysis has become increasingly prevalent in the evaluation of PV/T systems, as evidenced by recent literature ([2,15,16,26,32,33], and [34]). The calculations were conducted based on the specific weather conditions of Jahorm, Iran. The average amount of solar radiation in this location is between 2000 and 2200 kWh/m^2 per year, and this amount of radiation is suitable for solar power plants [35].

2. System explanation

A schematic diagram of the idea studied in this research is shown in Fig. 1(a). A system consisting of an SSCPV/T system (shown in Fig. 1(b)) and a heat exchanger has been used to provide electricity and hot water for a building located in Jahrom, Iran. Fig. 1 illustrates that the SSCPV/T system comprises two components: an optical filter and a PV/T receiver, both composed of multiple layers. The optical filter consists of three layers of glass, Ag/glycerol NF placed between the first and the second glass layers, and dotriacontane paraffin/gold NEPCM placed between the second and the third glass layers. By passing through this optical filter, a significant amount of the solar spectrum that is not in the visible range and therefore cannot be converted into electricity in the PV panel is separated from it. After leaving the optical filter and passing through an air layer, the sunlight reaches the receiver part, which consists of PV cells, an absorber, a coolant, and a back plate. The PV panel converts a significant portion of the absorbed sunlight into electricity. However, since not all absorbed radiation falls within the visible range, a portion of it transforms into heat, heating the PV panel. The absorber located under the panel takes this heat from the panel and transfers it into the water flow that passes under the absorber. The back plate also plays the role of insulation and prevents heat loss to the surrounding environment. The temperature of the optical fluid increases significantly and can be used to heat the water leaving the receiver further. For this purpose, a plate heat exchanger is used. After leaving the heat exchanger, the heated water enters a storage tank, and the optical fluid that is now cooled reenters the optical filter.

3. Mathematical modeling

To investigate the system's performance from the point of view of both the first and second laws of thermodynamics, the temporal changes in the temperature of different system parts are required. This involves formulating the energy conservation equations for these components and solving the system of governing equations, considering appropriate initial and boundary conditions. To write these equations, several assumptions have been included, which are as follows.

- ✓ For all components, heat transfer is done only in the vertical direction and in a time-dependent manner.
- ✓ PCM is homogeneous and isotopic, and its convection heat transfer mechanism is neglected.
- ✓ The heat exchange between the glass layers through the radiation mechanism is omitted.
- ✓ Except for NF and PCM, the thermophysical properties of other components remain constant and do not vary with temperature.
- ✓ The phase change of PCM results in changing its thermal and optical properties.
- ✓ The temperature of the optical filter as it exits the heat exchanger remains equal to its input temperature to the SSCPV/T system.

The energy balance equation for various parts of the SSCPV/T system is as follows.

- Glass 1 [32,33]:

$$\rho_{g1} c_{p,g1} \delta_{g1} \frac{\partial T_{g1}}{\partial t} = C\alpha_{g1} I_0 + \left(h_{c,g1-a} + h_{r,g1-s}\right)\left(T_a - T_{g1}\right) + h_{c,of-g1}\left(T_{of} - T_{g1}\right) \tag{1}$$

- Optical fluid [32,33]:

$$\rho_{of} c_{p,of} \delta_{of} \frac{\partial T_{of}}{\partial t} = C\tau_{g1}\alpha_{of} I_0 + h_{c,of-g1}\left(T_{g1} - T_{of}\right) + h_{c,of-g2}\left(T_{g2} - T_{of}\right) - \frac{\dot{m}_{of} c_{p,of}}{W} \frac{\partial T_{of}}{\partial y} \tag{2}$$

- Glass 2 [32,33]:

$$\rho_{g2} c_{p,g2} \delta_{g2} \frac{\partial T_{g2}}{\partial t} = C\tau_{g1}\tau_{of}\alpha_{g2} I_0 + h_{c,of-g2}\left(T_{of} - T_{g2}\right) + h_{cond,pcm-g2}\left(T_{pcm} - T_{g2}\right) \tag{3}$$

- PCM [36,37]:

$$\rho_{pcm}c_{p,pcm}\frac{\partial T_{pcm}}{\partial t}=k_{pcm}\frac{\partial^2 T_{pcm}}{\partial x^2}-\frac{\partial I}{\partial x} \tag{4}$$

- Glass 3 [32,33]:

$$\rho_{g3}c_{p,g3}\delta_{g3}\frac{\partial T_{g3}}{\partial t}=C\tau_{g1}\tau_{of}\tau_{g2}\tau_{pcm}\alpha_{g3}I_0+h_{cond,pcm-g3}\left(T_{pcm}-T_{g3}\right)+\left(h_{rad,pv-g3}+h_{conv,pv-g3}\right)\left(T_{pv}-T_{g3}\right) \tag{5}$$

- PV cell [32,33]:

$$\rho_{pv}c_{p,pv}\delta_{pv}\frac{\partial T_{pv}}{\partial t}=C\tau_{g1}\tau_{of}\tau_{g2}\tau_{pcm}\tau_{g3}\alpha_{pv}\left(1-\eta_{pv}\right)I_0+\left(h_{rad,pv-g3}+h_{conv,pv-g3}\right)\left(T_{g3}-T_{pv}\right)+h_{cond,pv-abs}\left(T_{abs}-T_{pv}\right) \tag{6}$$

- Absorber [32,34]:

$$\rho_{abs}c_{p,abs}\delta_{abs}\frac{\partial T_{abs}}{\partial t}=h_{cond,pv-abs}\left(T_{pv}-T_{abs}\right)+h_{c,cf-abs}\left(T_{cf}-T_{abs}\right) \tag{7}$$

- Coolant [32,34]:

$$\rho_{cf}c_{p,cf}\delta_{cf}\frac{\partial T_{cf}}{\partial t}=h_{c,cf-abs}\left(T_{abs}-T_{cf}\right)+h_{c,cf-bp}\left(T_{bp}-T_{cf}\right)-\frac{\dot{m}_{cf}c_{p,cf}}{W}\frac{\partial T_{cf}}{\partial y} \tag{8}$$

- Backplate [32,34]:

$$\rho_{bp}c_{p,bp}\delta_{bp}\frac{\partial T_{bp}}{\partial t}=h_{c,cf-bp}\left(T_{cf}-T_{bp}\right)+h_{c,bp-a}\left(T_a-T_{bp}\right) \tag{9}$$

where ρ, c_p, δ, T, C, α, I_0, τ, $\dot{m}$, W, k, h_c, h_{cond}, h_{conv}, and h_r, respectively stand for density, specific heat capacity, thickness, temperature, concentration ratio, absorptance, radiation intensity, transmittance, mass flow rate, width, thermal conductivity, combined convection-conduction heat transfer coefficient, conduction heat transfer coefficient, convection heat transfer coefficient, and radiation heat transfer coefficient. Moreover, the subscripts $g1$, $g2$, $g3$, a, s, of, pcm, pv, abs, cf, and bp, respectively stand for the first glass layer, second glass layer, third glass layer, ambient air, sky, optical filter, PCM, PV, absorber, coolant, and back plate.

The temperature of the water leaving the heat exchanger ($T_{cf,he}$) is also determined using the $\varepsilon-NTU$ method as follows [38]:

$$T_{cf,he}=T_{cf,o}+\frac{\varepsilon C_{min}\left(T_{of,o}-T_{cf,o}\right)}{\dot{m}_{cf}c_{p,cf}} \tag{10}$$

Where

$$\varepsilon=\frac{2}{1+\frac{C_{min}}{C_{max}}+\sqrt{1+\left(\frac{C_{min}}{C_{max}}\right)^2}+\frac{1+exp\left[-\frac{UA}{C_{min}}\sqrt{1+\left(\frac{C_{min}}{C_{max}}\right)^2}\right]}{1-exp\left[-\frac{UA}{C_{min}}\sqrt{1+\left(\frac{C_{min}}{C_{max}}\right)^2}\right]}} \tag{11}$$

$$C_{min}=min\left(\dot{m}_{of}c_{p,of},\dot{m}_{cf}c_{p,cf}\right) \tag{12}$$

$$C_{max}=max\left(\dot{m}_{of}c_{p,of},\dot{m}_{cf}c_{p,cf}\right) \tag{13}$$

Where ε, A and U, respectively stand for the effectiveness, heat transfer surface area of the heat exchanger, and overall heat transfer coefficient.

To solve the above equations to analyze the SSCPV/T system, a homemade MATLAB code based on the finite difference method with an implicit scheme for time marching was employed. The spatial differential terms were also discretized using the central difference method. The implicit scheme was selected due to its stability and accuracy in handling the governing equations. A time step of 0.5 s was chosen for the simulations, determined through a convergence analysis. Additionally, time steps of 0.1 and 1 s were tested to ensure the robustness and reliability of the results. The 0.5-s time step provided a balance between computational efficiency and accuracy. Some parameters in the relationships shown in this section have fixed values. The list of these parameters and their specified values is presented in Table 2.

3.1. Optical features of NEPCM and nanofluid

The characteristics of light scattering depend on the ratio of the particle size (r) to the wavelength of the incoming light (λ):

$$X = \frac{2\pi r}{\lambda} \tag{14}$$

To specify the optical characteristics of a spherical nanoparticle, the Mie scattering theory in $X \approx 1$ and Rayleigh scattering theory, which is a simplified approximation of Mie scattering theory, in $X \approx 0.1$ [41]. When using Rayleigh's theory, it is assumed that the particles are very far from each other, so that the light reflected from each particle does not hit the other particles.

In the following, the relations used to calculate the extinction coefficient for different states are presented.

- Single nanoparticle (Mie scattering theory [42]:

$$Q_{e\lambda} = \frac{2}{a^2}\sum_{n=1}^{\infty}(2n+1)Re(a_n + b_n) \tag{15}$$

$$a_n = \frac{\Psi_n(a)\Psi'_n\left(\frac{m_{particle}}{m_{medium}}a\right) - \frac{m_{particle}}{m_{medium}}\Psi'_n(a)\Psi_n\left(\frac{m_{particle}}{m_{medium}}a\right)}{\xi_n(a)\Psi'_n\left(\frac{m_{particle}}{m_{medium}}a\right) - \frac{m_{particle}}{m_{medium}}\xi'_n(a)\Psi_n\left(\frac{m_{particle}}{m_{medium}}a\right)} \tag{16}$$

$$Q_{e\lambda} = \frac{\frac{m_{particle}}{m_{medium}}\Psi_n(a)\Psi'_n\left(\frac{m_{particle}}{m_{medium}}a\right) - \Psi'_n(a)\Psi_n\left(\frac{m_{particle}}{m_{medium}}a\right)}{\frac{m_{particle}}{m_{medium}}\xi_n(a)\Psi'_n\left(\frac{m_{particle}}{m_{medium}}a\right) - \xi'_n(a)\Psi_n\left(\frac{m_{particle}}{m_{medium}}a\right)} \tag{17}$$

Where m is the complex refractive index, and Ψ_n and ξ_n are Riccati-Bessel functions.

- Single nanoparticle (Rayleigh scattering theory [42]:

$$Q_{e\lambda} = 4x\mathrm{Im}\left\{\frac{m^2-1}{m^2+2}\left[1 + \frac{x^2}{15}\left(\frac{m^2-1}{m^2+2}\right)\right]\right\} \tag{18}$$

Table 2
Fixed parameters employed in the simulations [39,40].

Parameter	Unit	Value	Parameter	Unit	Value
PV type	–	Polycrystalline silicon	$c_{p,PCM,l}$	J/kg.K	2428
$T_{cf,in}$	°C	25	$k_{PCM,s}$	W/m.k	0.6
C	–	30	$k_{PCM,l}$	W/m.K	0.154
δ_g	mm	3	δ_{pv}	mm	1
ρ_g	kg/m^3	2500	ρ_{pv}	kg/m^3	2320
$c_{p,g}$	J/kg.K	840	$c_{p,pv}$	J/kg.K	678
k_g	W/m.K	0.96	k_{pv}	W/m.K	100
ε_g	–	0.9	ε_{pv}	–	0.9
$T_{PCM,m}$	°C	70	α_{pv}	–	0.95
$T_{PCM,l}$	°C	70.5	δ_{abs}, δ_{bp}	mm	2
$T_{PCM,s}$	°C	69.5	ρ_{abs}, ρ_{bp}	kg/m^3	2698
h_{PCM}	J/kg	266000	$c_{p,abs}$, $c_{p,bp}$	J/kg.K	921
$\rho_{PCM,s}$	kg/m^3	809	k_{abs}, k_{bp}	W/m.K	226
$\rho_{PCM,l}$	kg/m^3	800	Pump efficiency	–	0.8
$c_{p,PCM,s}$	J/kg.K	1924			

- Dispersed particles group in the independent scattering regime [43]:

$$\beta_{np,\lambda} = \frac{3\phi_v Q_{e\lambda}}{2d_{np}} \tag{19}$$

Where ϕ_v is the volume fraction.

- Hybrid nanoparticles (NPs) [43]:

$$\beta_{np,\lambda} = \sum_{1}^{N} \beta_{np(i),\lambda} \tag{20}$$

- NEPCM [42]:

$$\beta_{NPCM,\lambda} = \beta_{np,\lambda} + \underbrace{\sigma_s + \frac{4\pi K_{PCM}}{\lambda}}_{\beta_{PCM,\lambda}} \tag{21}$$

where σ_s and K_{PCM}, respectively denote the scattering coefficient and extinction index.

To calculate the refractive and extinction coefficients of the PCM at various λ s, the absorptance and transmittance of NEPCM in the solid and liquid phases can be computed. When the PCM remains entirely pure and in the liquid phase, the scattering component of extinction coefficients becomes insignificant.

The optical properties of the liquid NEPCM are calculated as follows [44,45]:

$$\text{Transmittance } \tau_{\lambda,NEPCM} = \frac{\left(1 - r_{g-PCM}\right)\left(1 - r_{PCM-g}\right)\varsigma_{NEPCM}}{1 - r_{g-PCM} r_{PCM-g} \varsigma^2_{NEPCM}} \tag{22}$$

$$\text{Reflectance } r_{\lambda,NEPCM} = r_{g-PCM} + \frac{r_{PCM-g}\left(1 - r_{g-PCM}\right)^2 \varsigma^2_{NEPCM}}{1 - r_{g-PCM} r_{PCM-g} \varsigma^2_{NEPCM}} \tag{23}$$

$$\text{Absorptance } \alpha_{\lambda,NEPCM} = 1 - \tau_{\lambda,NEPCM} - r_{\lambda,NEPCM} \tag{24}$$

Where

$$r_{PCM-g} = \frac{\left(n_{PCM} - n_g\right)^2}{\left(n_{PCM} + n_g\right)^2} \tag{25}$$

$$r_{g-PCM} = \frac{\left(n_g - n_{PCM}\right)^2}{\left(n_g + n_{PCM}\right)^2} \tag{26}$$

$$\varsigma_{NEPCM} = e^{-\beta_{NEPCM,\lambda}\delta_{PCM}} \tag{27}$$

Where n is the refractive index.

The optical characteristics of the solid NEPCM are calculated as follows [46,47]:

$$\tau_{\lambda,NEPCM} = \frac{\left(1 - r_{g-PCM}\right)\left(1 - r_{PCM-g}\right)\varsigma_{NEPCM}}{1 - r_{g-PCM} r_{PCM-g} \varsigma^2_{NEPCM}} + \frac{1 - r_{d,g-PCM}}{1 - r_{d,g-PCM} r_{d,g-a} \varsigma^2_g} I_d(\chi, +1) \tag{28}$$

$$r_{\lambda,NEPCM} = r_{g-PCM} + \frac{r_{PCM-g}\left(1 - r_{g-PCM}\right)^2 \varsigma^2_{NEPCM}}{1 - r_{g-PCM} r_{PCM-g} \varsigma^2_{NEPCM}} + \frac{1 - r_{d,g-PCM}}{1 - r_{d,g-PCM} r_{d,g-a} \varsigma^2_g} I_d(\chi, +1) \tag{29}$$

$$\alpha_{\lambda,NEPCM} = 1 - \tau_{\lambda,NEPCM} - r_{\lambda,NEPCM} \tag{30}$$

Where

$$\chi = \beta_{NEPCM,\lambda}\delta_{NEPCM} \tag{31}$$

$$r_{d,i-j}=\frac{1}{2}+\frac{(m-1)(3m+1)}{6(m+1)^2}+\frac{m^2(m^2-1)^2}{(m^2+1)^3}ln\left(\frac{m-1}{m+1}\right)-\frac{2m^3(m^2+2m-1)}{(m^2+1)(m^4-1)}+\frac{8m^4(m^4+1)}{(m^2+1)(m^4-1)^2}ln(m) \tag{32}$$

$$I_d\left(\chi,\mu_j\right)=\frac{\omega\tau_g I_0\left(1-r_{g-PCM}\right)}{4}\left(1-\frac{\omega}{2}\sum_j\frac{w_j}{1-\mu_j/\mu_{PCM}}\right)\left[\frac{c_1}{1-\mu_j Z}e^{-Z\chi}+\frac{c_2}{1+\mu_j Z}e^{Z\chi}+\frac{e^{-\chi/\mu_{PCM}}}{1-\mu_j/\mu_{PCM}}\right] \tag{33}$$

Where m, ω, w_j, and μ_j, respectively denote the ratio of refractive index of two adjacent layers (>1), single scattering Albedo, quadrature weights, and direction cosines. In addition, c_1, and c_2 have a constant value calculated based on boundary conditions. Also, Z is computed as follows [46]:

$$Z=\frac{3}{2}\sqrt{\frac{1-\omega}{1-\omega/4}} \tag{34}$$

To calculate the total transmittance of NEPCM, it is necessary to integrate the transmission coefficient in the entire solar spectrum [33]:

$$\tau=\frac{\int_{250nm}^{2500nm}\tau_\lambda I_\lambda d\lambda}{\int_{250nm}^{2500nm}I_\lambda d\lambda} \tag{35}$$

The previously described procedure for computing the absorptance and transmittance coefficients of NEPCM can also be used for NF. However, in this case, the optical constants of NEPCM need to be substituted with those specific to the NF. It should be noted that the absorptance and transmittance coefficients of NF base fluid and PCM without NPs can also be calculated with the same relations, provided that the extinction coefficient of NPs in Eq. (21) should be removed.

3.2. Thermophysical properties of NEPCM and NF

The variation of the density and thermal conductivity of pure PCM via phase change is simplified considering linear distribution between solidus temperature ($T_{PCM,s}$) and liquidus temperature ($T_{PCM,l}$) of the PCM as follows [36,48]:

$$\rho=\begin{cases}\rho_s & T_{PCM}<T_{PCM,s}\\ \rho_s+(\rho_l-\rho_s)\left(T_{PCM}-T_{PCM,s}\right)/\left(T_{PCM,l}-T_{PCM,s}\right) & T_{PCM,s}<T_{PCM}<T_{PCM,l}\\ \rho_l & T_{PCM}>T_{PCM,l}\end{cases} \tag{36}$$

$$k=\begin{cases}k_s & T_{PCM}<T_{PCM,s}\\ k_s+(k_l-k_s)\left(T_{PCM}-T_{PCM,s}\right)/\left(T_{PCM,l}-T_{PCM,s}\right) & T_{PCM,s}<T_{PCM}<T_{PCM,l}\\ k_l & T_{PCM}>T_{PCM,l}\end{cases} \tag{37}$$

In this study, latent heat is considered using the apparent heat capacity method. In this method, the latent heat is included by raising the heat capacity of PCM during the phase change process [36,49]. A linear increase of the equivalent specific heat capacity from $c_{p,PCM,s}$ at $T_{PCM,l}$ to a maximum value of $c_{p,PCM,m}$ at $T_{PCM,m}$ and then a linear decrease to $c_{p,PCM,l}$ at $T_{PCM,l}$ through phase change is considered. Assuming the triangular area between these three points is equal to the latent heat (h_{PCM}), the maximum equivalent specific heat capacity ($c_{p,PCM,m}$) can be calculated as [36,49]:

$$c_{p,PCM,m}=c_{p,PCM,s}\frac{T_{PCM,l}-T_{PCM,m}}{T_{PCM,l}-T_{PCM,s}}+c_{p,PCM,l}\frac{T_{PCM,m}-T_{PCM,s}}{T_{PCM,l}-T_{PCM,s}}+\frac{2h_{PCM}}{T_{PCM,l}-T_{PCM,s}} \tag{38}$$

Therefore, by applying triangle similarity, the specific heat capacity can be evaluated as:

$$c_p=\begin{cases}c_{p,PCM,s} & T_{PCM}<T_{PCM,s}\\ c_{p,PCM,s}+\frac{c_{p,PCM,m}-c_{p,PCM,s}}{T_{PCM,m}-T_{PCM,s}}\left(T_{PCM}-T_{PCM,s}\right) & T_{PCM,s}<T_{PCM}<T_{PCM,m}\\ c_{p,PCM,l}+\frac{c_{p,PCM,m}-c_{p,PCM,l}}{T_{PCM,l}-T_{PCM,m}}\left(T_{PCM,l}-T_{PCM}\right) & T_{PCM,m}<T_{PCM}<T_{PCM,l}\\ c_{p,PCM,l} & T_{PCM}>T_{PCM,l}\end{cases} \tag{39}$$

The thermophysical features of glycerol can be obtained as follows [50]:

$$\rho = 349.08 \times 0.24902^{-\left(1-\frac{T}{723}\right)^{0.15410}} \tag{40}$$

$$\log_{10} k = -0.3550 - 0.2097\left(1 - \frac{T}{723}\right)^{\frac{2}{7}} \tag{41}$$

$$c_p = 1434.8954 + 9.3390T - 0.0214T^2 + 0.0196 \times 10^{-3}T^3 \tag{42}$$

$$\mu = \left(6.092563 \times 10^{-22}\right) \times 10^{\left(\frac{4230.5}{T} + 0.028705T - 1.8648 \times 10^{-5}T^2\right)} \tag{43}$$

4. Calculation of performance metrics

The system under investigation in the current research can convert the energy in the irradiated sunlight into thermal energy, electrical energy, thermal exergy, or electrical exergy, and the success rate of the system in this task can be measured by using first-law thermal efficiency (η_{th}), first-law electrical efficiency (η_{el}), second-law thermal efficiency (σ_{th}), and second-law electrical efficiency (σ_{el}), respectively, calculating as follows [26,34]:

$$\eta_{th} = \frac{\dot{m}_{cf} c_{p,cf}\left(T_{cf,he} - T_{cf,in}\right)}{CI_0 WL} \tag{44}$$

$$\eta_{el} = \eta_{pv} - \frac{P_{pump}}{CI_0 WL} \tag{45}$$

$$\sigma_{th} = \frac{\dot{m}_{cf} c_{p,cf}\left[\left(T_{cf,he} - T_{cf,in}\right) - T_a \ln\left(\frac{T_{cf,h}}{T_{cf,in}}\right)\right]}{CI_0 WL\left[1 - \frac{4}{3}\frac{T_a}{T_{sun}} + \frac{1}{3}\left(\frac{T_a}{T_{sun}}\right)^4\right]} \tag{46}$$

$$\sigma_{el} = \frac{CI_0 A\eta_{pv} - P_{pump}}{CI_0 WL\left[1 - \frac{4}{3}\frac{T_a}{T_{sun}} + \frac{1}{3}\left(\frac{T_a}{T_{sun}}\right)^4\right]} \tag{47}$$

Where L is the length of the CPV/T unit. In the aforementioned equations, T_{sun} represents the temperature of the sun, typically considered as 5800 K.

The overall performance of the system from the viewpoint can be measured based on the overall first law (η_{ov}), and second law (σ_{ov}) efficiencies, which are calculated as follows [30]:

$$\eta_{ov} = \eta_{th} + \eta_{el}/0.38 \tag{48}$$

$$\sigma_{ov} = \sigma_{th} + \sigma_{el} \tag{49}$$

The definition of the overall first law efficiency considers the higher quality and value of electrical energy compared to thermal energy, in line with the second law of thermodynamics. By dividing the electrical efficiency by the efficiency of a typical thermal power plant (38 %), we enhance the contribution of electricity to the overall efficiency. As a result of this definition, overall efficiencies exceeding 100 % are possible and have been observed in other studies, such as Zhang et al. [30]. This approach allows for a more comprehensive representation of the system's performance by acknowledging the superior quality of electrical energy.

5. Results and discussions

In this section, the homemade MATLAB code prepared to examine the performance of the SSCPV/T unit is first validated. Then, this code is used to check the SSCPV/T system performance. For this purpose, first, the performance of the system on a specific day, i.e., 21st June, is investigated. June 21st is an important date in meteorological studies because it marks the summer solstice, which is the longest day of the year. This day provides maximum solar irradiance, making it a crucial time for evaluating system performance under peak conditions. Then, the effect of L (1–9 m), $\dot{m}_{of}$ (0.005–0.5 kg/s), $\dot{m}_{cf}$ (0.005–0.5 kg/s), UA (25–125 W/K), δ_{of} (5–20 mm), δ_{cf} (5–20 mm), δ_{pcm} (5–20 mm), and δ_{abs} (5–20 mm) on the annual average thermal, electrical, and overall energy and exergy efficiencies of the system are examined. When checking the hourly and daily performance of the system, the considered values for the influential variables are $L = 1$ m, $\dot{m}_{of} = \dot{m}_{cf} = 0.05$ kg/s, $UA = 125$ W/K, and $\delta_{of} = \delta_{cf} = \delta_{pcm} = \delta_{abs} = 10$ mm.

5.1. Validation

To assess the validity of the modeling done in this research, the data provided by Yazdanifard [26] for the hourly changes in the PV panel temperature of an SSCPV/T unit was employed. In the considered case, the thickness of the NEPCM layer is 10 mm. The output of this comparison is presented in Fig. 2. It is evident that the difference between the results of the two research is insignificant, and the model used in the present study is valid. It should be noted that in Ref. [26], the models presented for predicting the optical properties of pure PCM, NEPCM, and NF have also been validated. This has been avoided in the present research because the validation of the current research has been done by applying the results presented in Ref. [26]. Confirming the validity of the entire modeling process inherently confirms the validity of each constituent part, encompassing the models used for predicting the optical properties of various system components.

5.2. Assessing the hourly and daily performance of the system

Time changes in solar radiation intensity, ambient temperature, and wind velocity of Jahorm city on June 21st are shown in Fig. 3. These data were extracted from the website of the Iran Meteorological Organization [35].

Fig. 4 displays the temperature changes of the PV panel (T_{pv}), PCM, NF (i.e., spectral splitter) (T_{of}), water leaving the SSCPV/T unit ($T_{cf,out}$), and water leaving the heat exchanger ($T_{cf,he}$) during June 21st. The change range of these temperatures is T_{pv} = 297.79–319.47 °C, T_{PCM} = 297.78–438.15 °C, T_{of} = 297.87–377.26 °C, $T_{cf,out}$ = 298–301.44 °C, and $T_{cf,he}$ = 298–306.55 °C. The highest value recorded for all these temperatures occurs between 12:30 to 13:00, which is predictable considering that the intensity of solar radiation is maximum in this period. The PCM layer experiences the highest temperature as it is stationary and is only cooled from the upper layer (NF), lacking adequate cooling from the lower layer (air gap). In Fig. 4, two regions with constant temperatures (at about $T_{PCM,m}$ = 343) can be seen in the PCM layer for about an hour in the morning and afternoon exhibiting the charging and discharging cycles of the PCM. These stability regions indicate the transitions in the phase of PCM: from solid to liquid during charging, and from liquid to solid during discharging. Due to the presence of various interactive thermal components in the proposed system, the stabilizing thermal effects (constant temperature region) of the phase change material (PCM) cannot be directly observed in the other parts like PV, NF, and CF. In similar studies [34,47,48], the constant temperature regions are also all specific to the PCM. In addition, the temperature of the NF has reached 377.26 K during some hours of the day, which shows that by changing the functional parameters of the system, the rate of heat transfer from the NF to the domestic water, and thus the efficiency of the system can be improved. Moreover, according to the results, the average daily η_{th}, η_{el}, η_{ov}, σ_{th}, σ_{el} and σ_{ov} for June 21st are 0.584, 0.112, 0.878, 0.140, 0.120 and 0.134, respectively.

5.3. Sensitivity analysis of system parameters

The annual average effects of various parameters on the performance of the SSCPV/T system have been investigated. These parameters include the system's length, the overall heat transfer coefficient of the heat exchanger, the thickness of the phase change material (PCM), the spectral splitter, as well as the layers of coolant and air gaps. Additionally, the mass flow rate of the spectral splitter and water has been considered. To evaluate the system's annual performance, the time variation of different parameters was first assessed on the 21st day of each month throughout the year. The annual behavior of the system was then analyzed by calculating the time average of the collected data. How the length of the SSCPV/T unit (L) affects the performance of the SSCPV/T-heat exchanger system is indicated in Fig. 5. It is evident in Fig. 5(a) that as the L increases, both T_{pv} and $T_{cf,he}$ increase, the former being unfavorable, and the latter being favorable. As the L increases from 1 m to 9 m, the T_{pv} increases from 305.1 K to 307.7 K, and the $T_{cf,he}$ augments from 301.9 K to 317.1 K. As L rises, the PV panel absorbs more solar energy, consequently raising the T_{pv}. Additionally, raising L results in an extended channel length for water flow. The water enters the channel with a temperature of 25 °C, which gradually increases after receiving heat from the PV panel. Therefore, the water at the beginning of the channel has a much better cooling ability than the

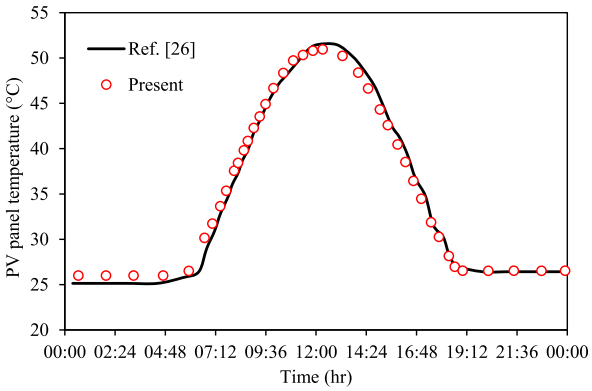


Fig. 2. Findings of validation versus Ref. [26].

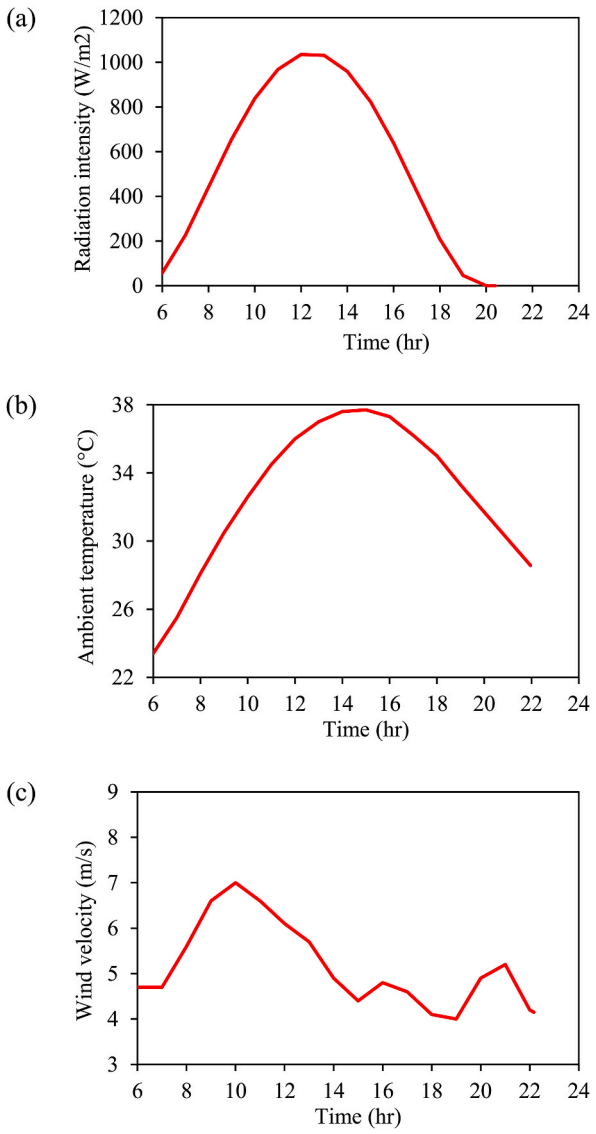


Fig. 3. Weather data of Jahrom city on June 21st, (a) solar radiation intensity, (b) ambient temperature, and (c) wind velocity [35].

end of the channel. As a result, the shorter the channel, the better its cooling. The combination of these two factors causes the T_{pv} to increase as the L increases. In addition, increasing the L means increasing the heat transfer surface area from the panel to the water, which results in an increase in the $T_{cf.he}$. Fig. 5(b) shows that as the L increases, the η_{th} and η_{ov} drop significantly while the η_{el} increases slightly. As the L increases from 1 m to 9 m, the η_{th} decreases from 71.56 % to 39.23 %, and the η_{ov} decreases from 99.97 % to 68.4 %, while the η_{el} boosts from 10.79 % to 11.08 %. With the increase in L, both the useful thermal power yield of the system and the solar energy obtained by the system boost. The pattern of results evident in Fig. 5(b) reveals that the increase in useful thermal power is less than the increase in solar radiation hitting the system, which has caused the η_{th} to decrease with increasing L. Changes in the L affect the η_{el} in three different ways, which are: (i) An increase in the T_{pv} due to the increase in L is associated with a decrease in the PV panel electrical efficiency and, as a result, a decrease in the η_{el}; (ii) Increasing the L causes an increase in the pressure drop within the channel, necessitating more pumping power for water flow and subsequently reducing the η_{el}.; (iii) An increase in solar power absorbed by the system, which leads to an increase in the η_{el}. Combining these three factors has caused the η_{el} to remain almost constant with increasing L. The increase in L is associated with a significant decrease in the η_{th}, and the η_{el} remains constant, which has caused the η_{ov} to follow the pattern of η_{th} and decrease drastically with the increase in L. Fig. 5(c) also reveals that the σ_{th}, σ_{el} and σ_{ov} elevate with increasing L. As the length L from 1 m to 9 m, σ_{th}, σ_{el} and σ_{ov}, respectively augment from 1.12 % to 2.44 %, from 11.56 % to 11.87 %, and from 12.68 % to 14.30 %. By comparing the results presented in Fig. 5(a) and 5(b), we can see the difference between the quality of thermal and electrical energies. In Fig. 5, the value of η_{el} is always lower than the η_{th}, but the opposite of this observation is valid for σ_{el}.

UA in the context of a heat exchanger refers to the product of the overall heat transfer coefficient (U) and the heat transfer surface

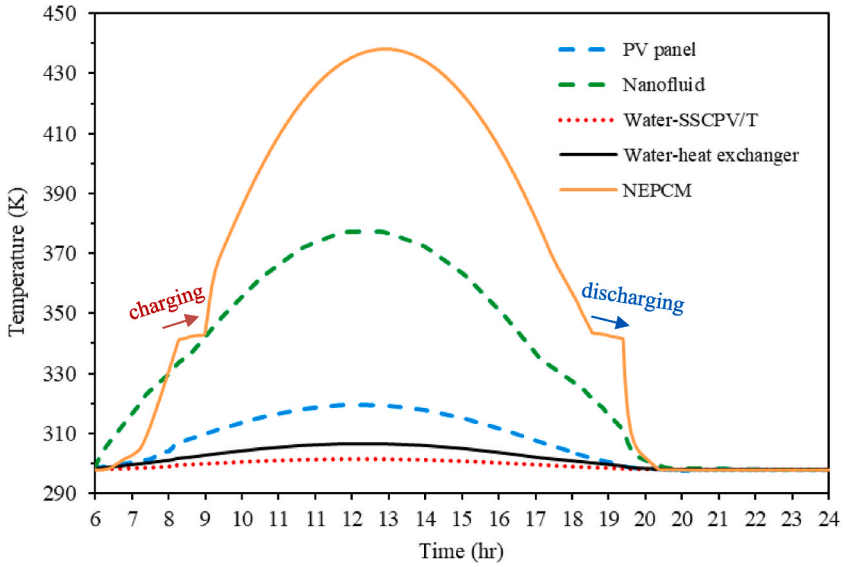


Fig. 4. Time variation of the temperatures of PV panel, NF, PCM, and water leaving the SSCPV/T unit and heat exchanger on June 21st.

area (A). UA, therefore, gives you an overall measure of the heat exchanger's capacity to transfer heat. Fig. 6 illustrates the impact of UA on the performance of the SSCPV/T-heat exchanger system. Fig. 6(a) reveals that as the UA increases, the T_{pv} remains constant, but the $T_{cf,he}$ augments slightly (from 301.59 to 301.86 K). Considering that water enters the heat exchanger after leaving the SSCPV/T system, it was expected that changing the characteristics of the exchanger would not affect the T_{pv}. In addition, as depicted in Fig. 6(b), the rise of UA is associated with the increase of η_{th} (from 66.38 % to 71.56 %), and η_{ov} (from 95.36 % to 99.96 %), and the decrease of η_{el} (from 11.01 % to 10.79 %). The increase in η_{th} with the rise of UA is attributed to the expansion of the useful heat acquired by the water within the heat exchanger, while the decrease in η_{el} is due to the increase in the pressure drop of the water inside the heat exchanger, which leads to a rise in the pumping power. The increase in η_{ov} with increasing UA also reveals that the rise in η_{th} with increasing UA is more significant than the decrease in η_{el}. It is also evident in Fig. 6(c) that with increasing UA, the σ_{th} increases slightly from 0.956 % to 1.124 %, while σ_{el} decrease from 11.79 % to 11.56 % and σ_{ov} decreases from 12.75 % to 12.68 %.

Fig. 7 gives the consequences of boosting the $\dot{m}_{of}$ on the performance metrics of the SSCPV/T-heat exchanger system. As seen in Fig. 7(a), with the increase in the $\dot{m}_{of}$, the T_{pv} and $T_{cf,he}$ decrease from 305.41 K to 304.91 K and increase from 301.03 K to 302.14 K, respectively, which are both favorable observations. Increasing the $\dot{m}_{of}$ enhances the spectral splitter's performance, resulting in an elevation of the NF temperature and a reduction in T_{pv}. Moreover, raising the NF temperature exiting the SSCPV/T system increases the heat transferred from the NF to the water within the heat exchanger. In addition, the consequence of increasing the $\dot{m}_{of}$ from 0.005 kg/s to 0.5 kg/s is an increase in the η_{th} from 55.92 % to 77.07 %, a decrease in the η_{el} from 11.04 to 6.47 %, and an ascending-descending trend of η_{ov}. The highest η_{ov} equals 101.4 %, which occurs at $\dot{m}_{of} = 0.1$ kg/s. The improvement in the η_{th} is a result of the increase in $T_{cf,he}$, while the decrease in the η_{el} is due to the rise in the pressure drop of the NF flow due to the rise in the $\dot{m}_{of}$. Changes in the η_{ov} also show that up to the $\dot{m}_{of}$ of 0.1 kg/s, the increasing effect of η_{th} on η_{ov} has overcome the decreasing effect of η_{el}, conversely, for higher $\dot{m}_{of}$, the situation is reversed. Fig. 7(c) also shows that the pattern of changes in σ_{th}, σ_{el} and σ_{ov} with increasing $\dot{m}_{of}$ is ascending (from 0.696 % to 1.31 %), descending (from 11.82 % to 6.93 %), and ascending-descending with the maximum occurring at $\dot{m}_{of} = 0.05$ kg/s.

Fig. 8 illustrates how the performance of the SSCPV/T-heat exchanger system is impacted by changes in $\dot{m}_{cf}$. When the $\dot{m}_{cf}$ increases up to 0.1 kg/s, both the T_{pv} and the $T_{cf,he}$ fall drastically, and this downward trend is also observed at higher flow rates, but with less intensity. With an increase in $\dot{m}_{cf}$ from 0.005 to 0.5 kg/s, there's a decline in T_{pv} from 317.18 to 302.03 K and a decrease in $T_{cf,he}$ from 329.56 to 298.40 K. The increase in $\dot{m}_{cf}$ causes an elevation in the convection heat transfer coefficient, leading to increased heat transfer from the PV panel to the water flow, subsequently resulting in the cooling of the PV panel. On the other hand, when the $\dot{m}_{cf}$ increases, the time that the water is in contact with the PV panel and heat exchanger surfaces decreases, leading to a subsequent decrease in $T_{cf,he}$. In Fig. 8(b), it is evident that increased $\dot{m}_{cf}$ is associated with the increase in η_{th} and η_{ov} and the decrease in η_{el}. As the $\dot{m}_{cf}$ increases from 0.005 kg/s to 0.5 kg/s, η_{th} increases from 58.39 % to 73.57 %, η_{el} decreases from 10.69 % to 10.55 % and η_{ov} increases from 86.52 % to 101.32 %. The increase in η_{th} is due to the increase in heat transfer from the PV panel and NF to water. With the increase in $\dot{m}_{cf}$, due to the decrease in the T_{pv}, the electrical efficiency of the panel and as a result, the η_{el} improves, but on the other hand, the pumping power required to circulate water into the SSCPV/T system and heat exchanger increases. The results presented in Fig. 8(b) reveal that the effect of increasing pumping power on the η_{el} overcomes the effect of increasing the PV panel efficiency, and, as a result, η_{el} decreases with increasing $\dot{m}_{cf}$. The upward trend of η_{ov} with increasing $\dot{m}_{cf}$ also reveals that the increase in η_{th} with increasing $\dot{m}_{cf}$ is more significant than the decrease in η_{el}. Fig. 8(c) also shows a significant decrease in σ_{th} (from 5.64 % to 0.313 %), and σ_{ov} (from 17.08 % to 11.61 %) with increasing $\dot{m}_{cf}$ from 0.005 kg/s to 0.5 kg/s, while the pattern of changes in σ_{el} is fluctuating so

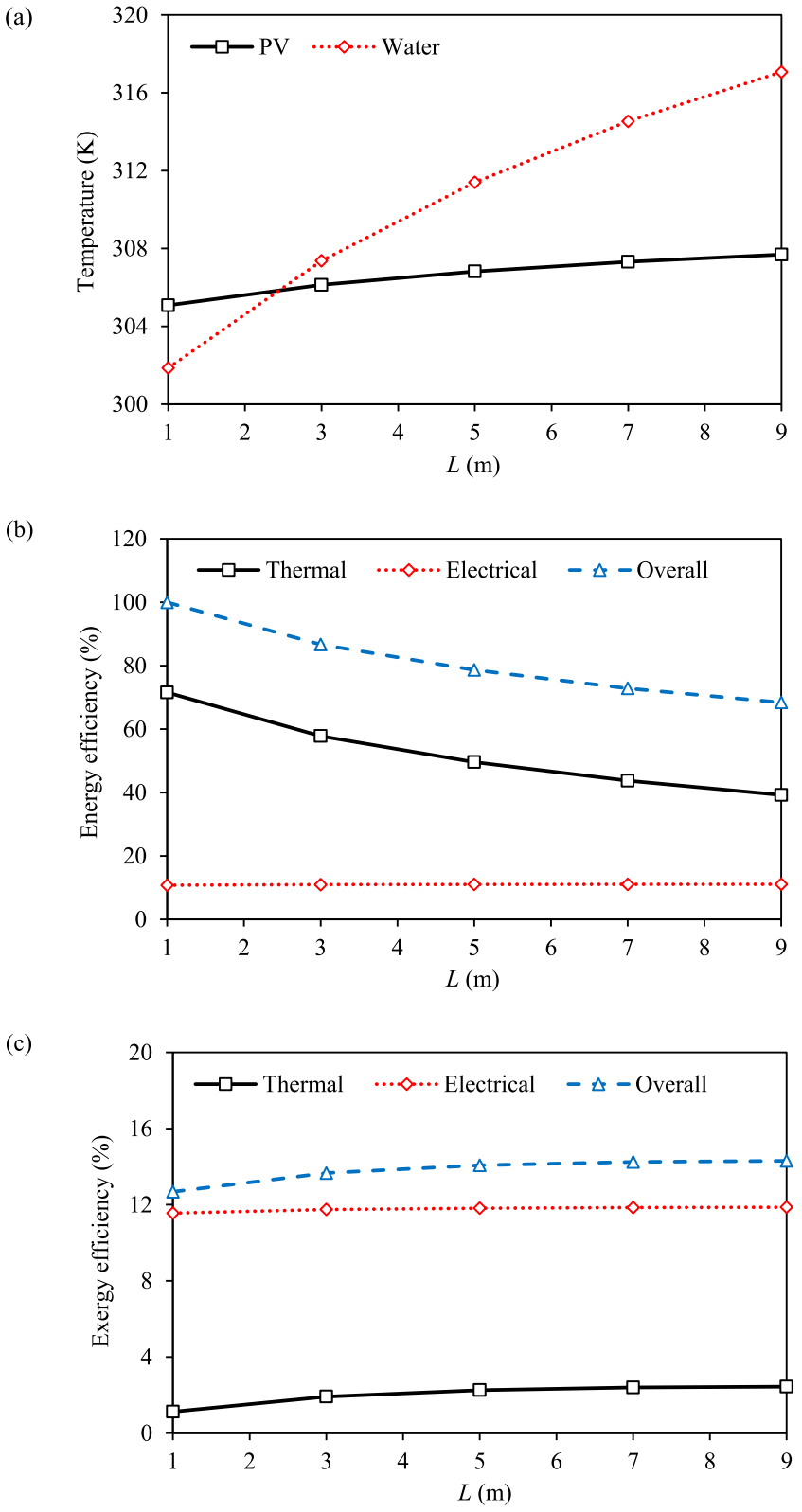


Fig. 5. Impact of L on the performance metrics of the hybrid system.

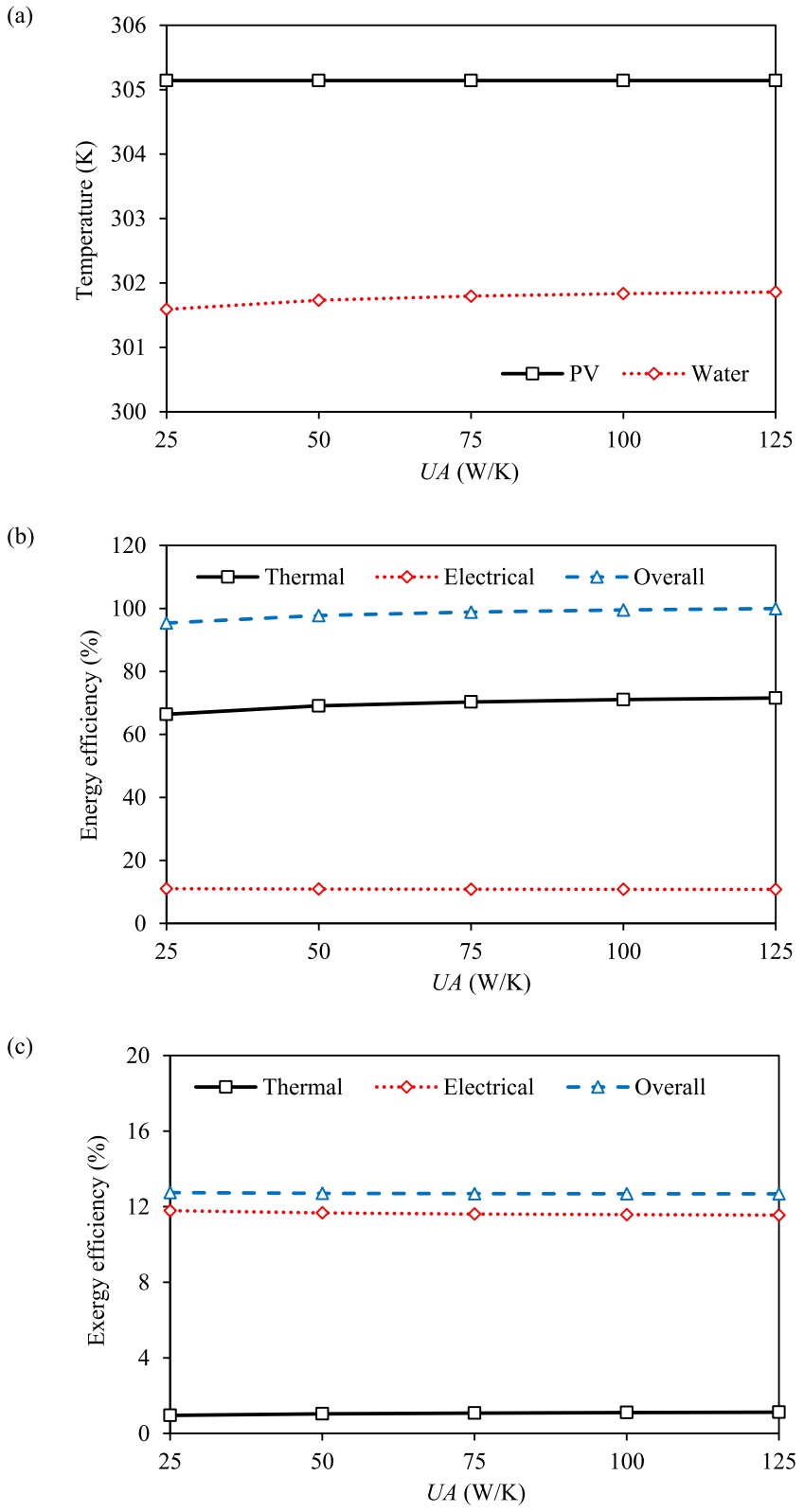


Fig. 6. Impact of *UA* on the performance metrics of the hybrid system.

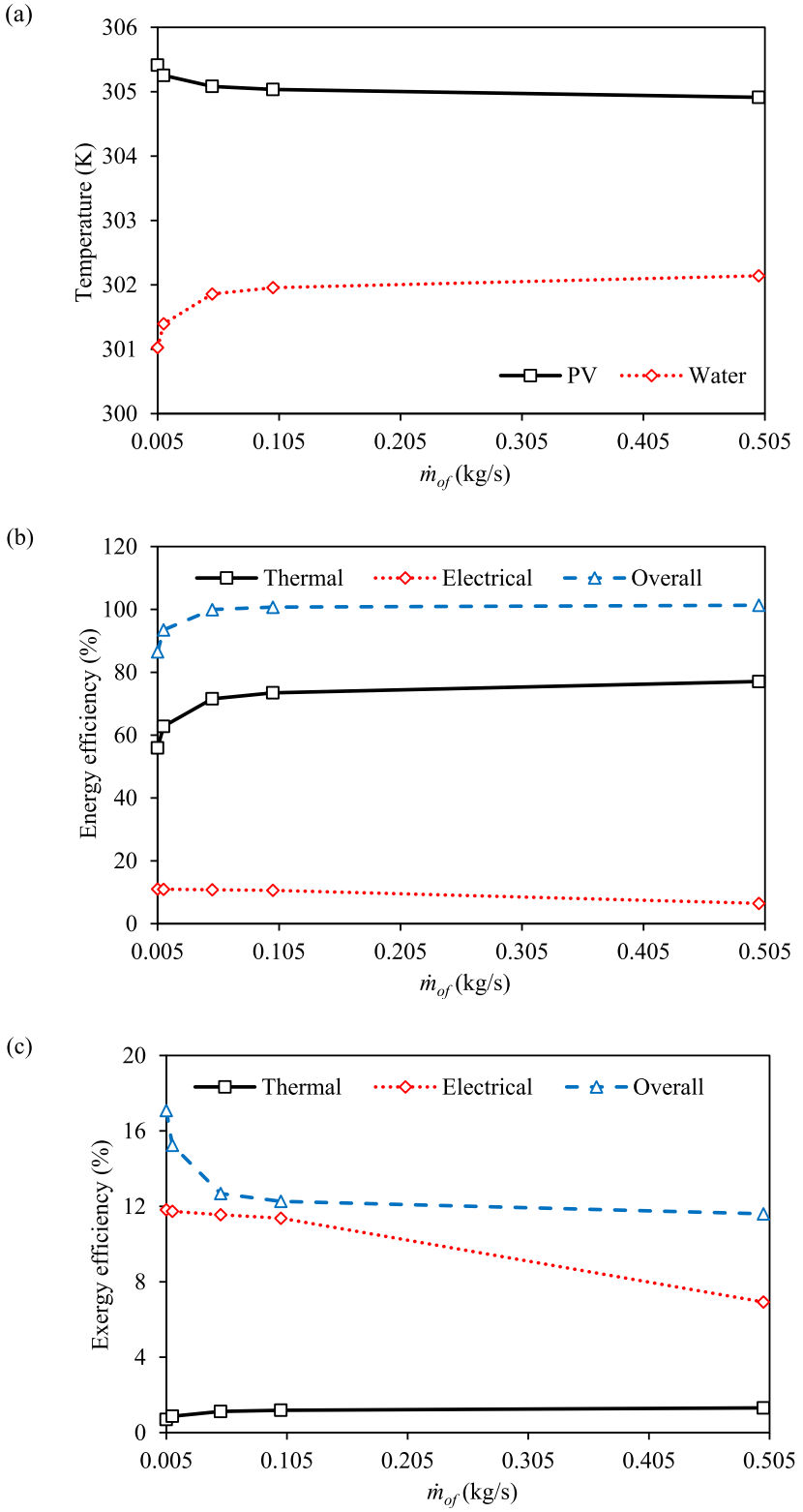


Fig. 7. The influence of $\dot{m}_{of}$ on the performance metrics of the hybrid system.

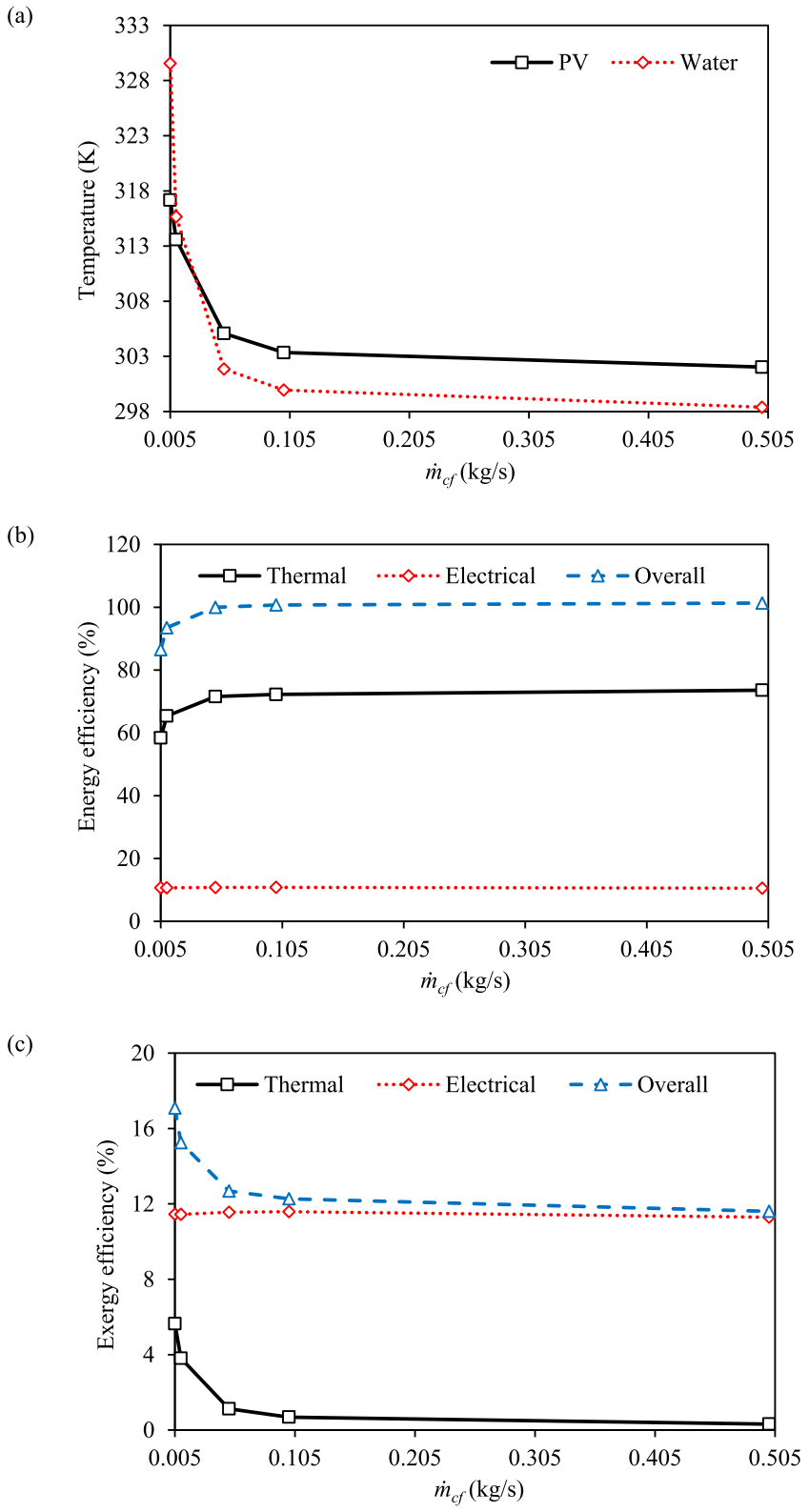


Fig. 8. Impact of $\dot{m}_{cf}$ on the performance metrics of the hybrid system.

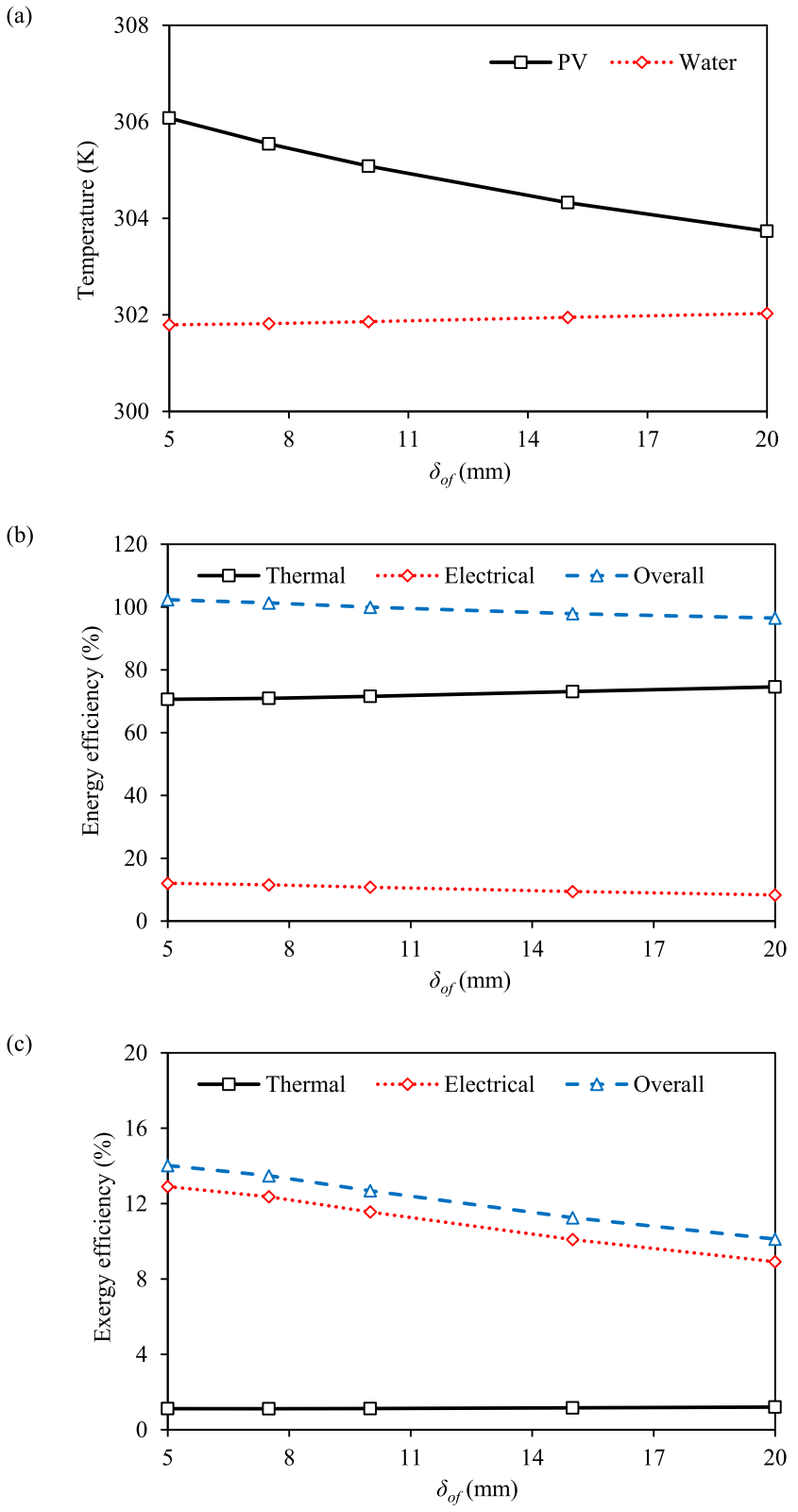


Fig. 9. Impact of δ_{of} on the performance metrics of the hybrid system.

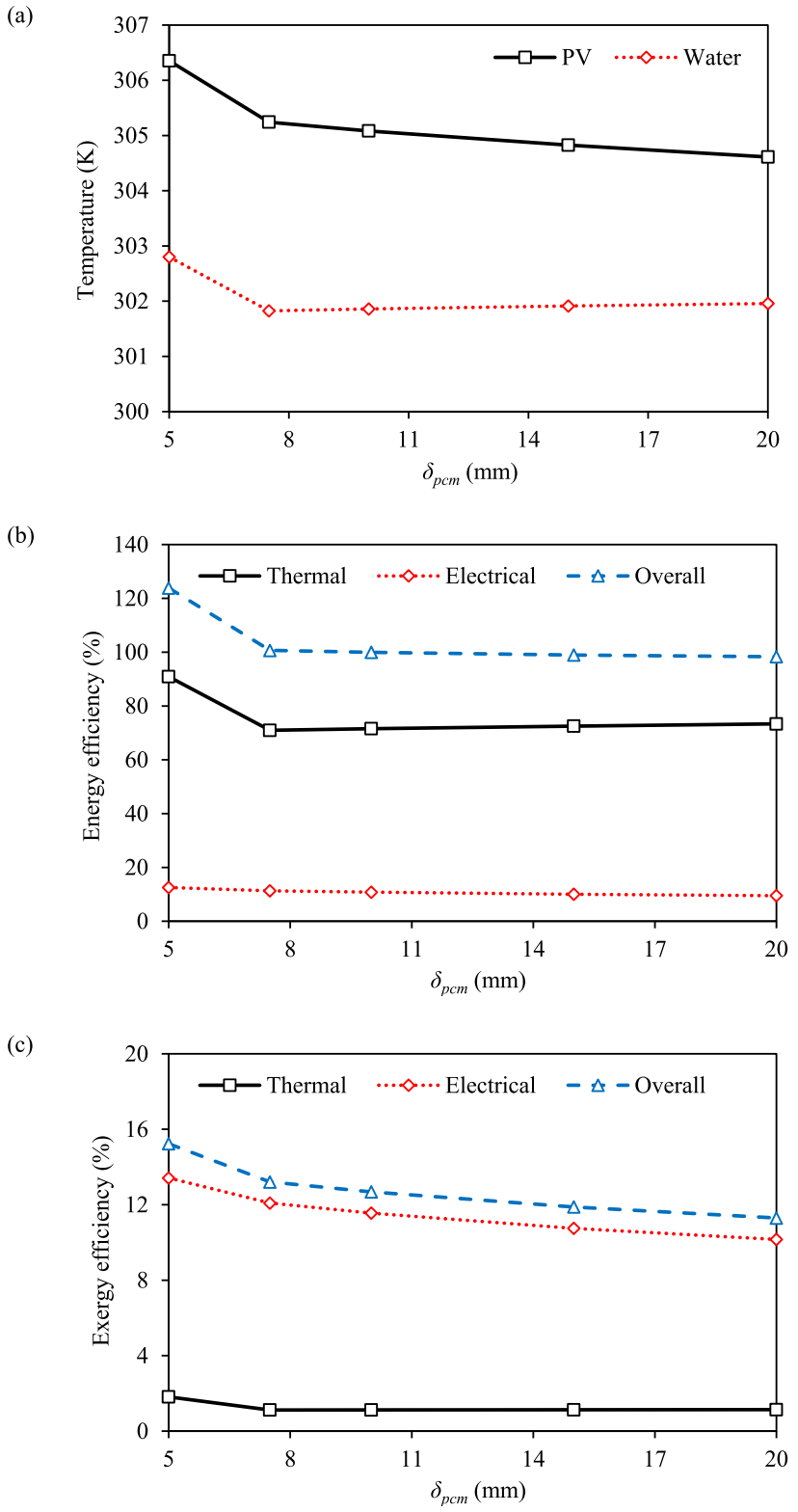


Fig. 10. Impact of δ_{pcm} on the performance metrics of the hybrid system.

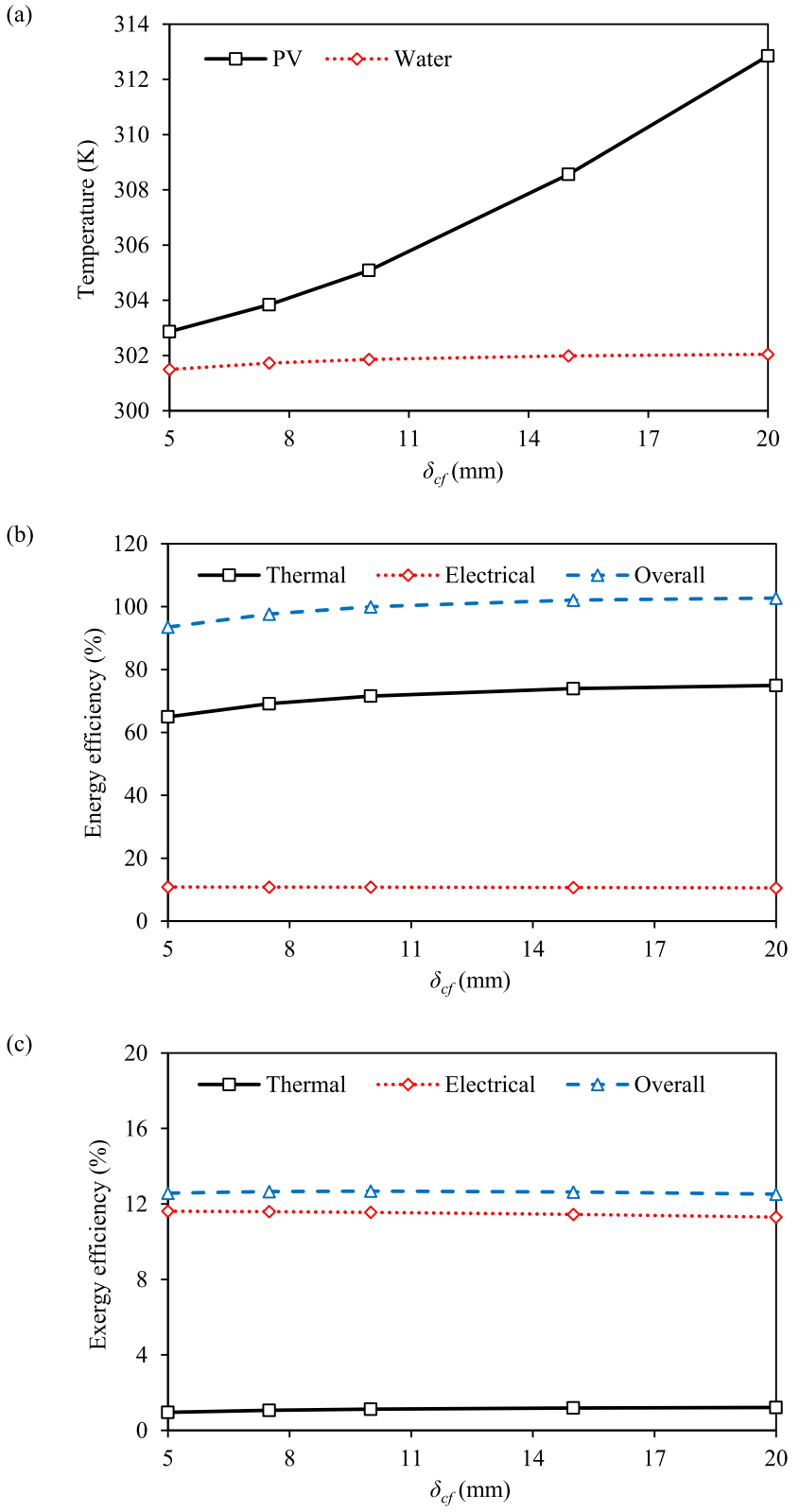


Fig. 11. Impact of δ_{cf} on the performance metrics of the hybrid system.

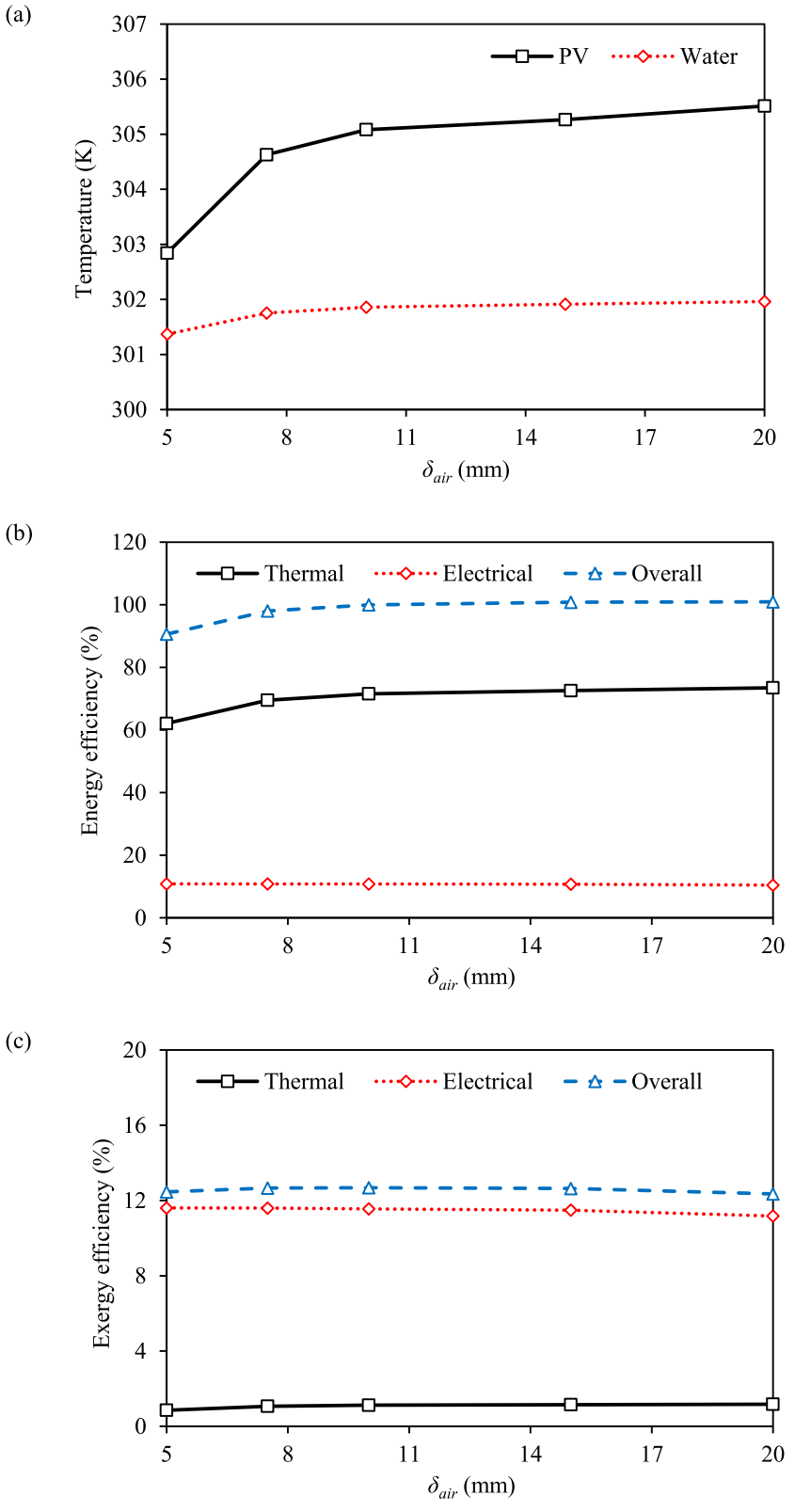


Fig. 12. Impact of δ_{air} on the performance metrics of the hybrid system.

that with increasing $\dot{m}_{cf}$ from 0.005 kg/s to 0.01 kg/s and 0.1 kg/s to 0.5 kg/s, it decreases, while increasing $\dot{m}_{cf}$ from 0.01 kg/s to 0.1 kg/s is associated with increased σ_{el}.

Fig. 9 reveals how the δ_{of} change affects the performance of the hybrid system. Fig. 9(a) depicts that as the δ_{of} increases from 5 mm to 20 mm, the T_{pv} decreases from 306.08 K to 303.73 K, and the $T_{cf,he}$ increases from 301.79 K to 302.03 K. Increasing the δ_{of} leads to a rise in the amount of radiation absorbed by this layer, resulting in less radiation passing through it. This effect causes a decrease in the T_{pv}, meaning the temperature of the fluid at the PV panel drops. As the δ_{of} increases, the absorbed radiation further contributes to a reduction in the radiation reaching the PV panel, which in turn lowers the temperature of the water exiting the SSCPV/T system. Additionally, the increased absorption of radiation by the nanofluid (NF) results in a higher temperature of the NF leaving the SSCPV/T system. This increase enhances the rate of heat transfer from the NF to the water flow within the heat exchanger. Consequently, while raising δ_{of} decreases T_{pv} and reduces the heat transfer rate from the PV panel to the cooling water, it simultaneously increases the temperature of the nanofluid, thereby enhancing the heat transfer rate from the NF to the water in the heat exchanger. The results presented in Fig. 9(a) reveal that the increasing effect of δ_{of} on $T_{cf,he}$ is more effective than its decreasing effect. Fig. 9(b) reveals that the increase in δ_{of} from 5 mm to 20 mm is accompanied by a decrease in the η_{el} (from 12.05 % to 8.33 %), and η_{ov} (from 102.33 % to 96.49 %), and an increase in the η_{th} (from 70.62 % to 74.57 %). The increasing trend of the η_{th} and the decreasing trend of the η_{el} can be related to the increase in the $T_{cf,he}$ and the decrease in the radiation received by the PV panel. It should be noted that because the $\dot{m}_{cf}$ is considered constant when the δ_{of} is changed, it can be claimed that the NF velocity and as a result, the pressure drop and the pumping power of the NF decrease with the increase in δ_{of}, which leads to an increase in the η_{el}. However, the effect of reducing the radiation received by the PV panel on the η_{el} has overcome the effect of reducing the pumping power. The results of η_{ov} also show that despite the increase in η_{th}, the increase in δ_{of} is not desirable at all from the overall energy point of view, because it severely reduces the electrical power produced by the system. It can also be seen in Fig. 9(c) that with the rise of δ_{of} from 5 mm to 20 mm, the σ_{el} decreases from 12.90 % to 8.92 % and the σ_{ov} decreases from 14.02 % to 10.12 %, but the σ_{th} increases from 1.12 % to 1.20 % with increasing δ_{of} from 5 mm to 7.5 mm, but it follows an upward trend at higher δ_{of}.

The consequences of increasing the δ_{pcm} on the performance features of the SSCPV/T-heat exchanger system are shown in Fig. 10. It can be seen in Fig. 10(a) that the increase in δ_{pcm} is accompanied by a decrease in the T_{pv} from 306.35 K to 304.61 K, which is caused by a decrease in the radiation received by the PV panel. Also, Fig. 10(a) reveals that with the increase in δ_{pcm} from 5 mm to 7.5 mm, the $T_{cf,he}$ decreases from 302.80 K to 301.82 K, while a further increase in the δ_{pcm} is associated with a slight increase in the $T_{cf,he}$ from 301.82 K to 301.96 K. As the T_{pv} decreases, the heat transfer rate from the PV panel to the water flow passing under the PV panel decreases, leading to a reduction in the temperature of the water leaving the SSCPV/T system. When water enters the heat exchanger with a lower temperature, it can take more heat from the NF flow, which means that the $T_{cf,he}$ increases with the increase in the δ_{pcm}. Analyzing the results presented in Fig. 10(a), it can be seen that up to the δ_{pcm} of 7.5 mm, the decreasing effect of δ_{pcm} on the $T_{cf,he}$ is more significant than its increasing effect, while the opposite of this observation is true for larger δ_{pcm} s. In Fig. 10(b), it can be observed the pattern of variations in η_{th}, η_{el}, and η_{ov} with respect to δ_{pcm}. The downward trend of η_{el} (from 12.53 % to 9.49 %), and η_{ov} (from 123.87 % to 98.31 %), and the downward-ascending trend of η_{th} with increasing δ_{pcm} from 5 mm to 20 mm are evident in this figure. The decrease in the η_{el} is caused by the decrease in the PV panel temperature, while the pattern of changes in η_{th} follows the pattern of $T_{cf,he}$. Also, the decrease of the η_{ov} is a sign of a more significant impact of the η_{el} compared to the η_{th}. Analyzing the results given in Fig. 10(c) also exhibits the similarity of the changing pattern of σ_{th}, σ_{el} and, σ_{ov} with that of η_{th}, η_{el}, and η_{ov}, respectively. As the δ_{pcm} augments from 5 mm to 20 mm, σ_{el} declines from 13.42 % to 10.16 %, σ_{ov} declines from 15.23 % to 11.30 %, and σ_{th} first diminishes from 1.82 % to 1.12 % (at $\delta_{pcm} = 7.5$ mm), and then, increases from 1.12 % to 1.14 %.

Fig. 11 illustrates how the performance of the SSCPV/T-heat exchanger system is affected by δ_{cf}. A significant unfavorable rise in the T_{pv} (from 302.87 K to 312.85 K), and a slight favorable rise in the $T_{cf,he}$ (from 301.50 K to 302.04 K) with increasing δ_{cf} can be seen in Fig. 11(a). If the water mass flow rate is constant, increasing the δ_{cf} means decreasing the water velocity, which results in, firstly reducing the cooling ability of water, and, as a result, increasing the T_{pv} and, secondly, increasing the $T_{cf,he}$. Fig. 11(b) also reveals that the enhancement in the δ_{cf} is correlated with a rise in the η_{th} (from 64.95 % to 74.94 %), and η_{ov} (from 93.50 % to 102.73 %), and a decrease in the η_{el} (from 10.85 % to 10.56 %). The increase, decrease, and increase of η_{th}, η_{el} and, η_{ov} with boosting δ_{cf}, respectively, are caused by the increase in the $T_{cf,he}$, increase in the T_{pv}, and the dominance of the increasing effect of η_{th} on the η_{ov} over the decreasing effect of η_{el}. It can also be seen in Fig. 11(c) that the increase in δ_{cf} is associated with a decrease in the σ_{el} (from 11.62 % to 11.31 %), an increase in the σ_{th} (from 0.956 % to 1.21 %), and an increase-decrease in the σ_{ov}. The maximum σ_{ov} equals 12.68 %, which occurs at $\delta_{cf} = 10$ mm.

Fig. 12 demonstrates the impact of the thickness of the air gap (δ_{air}) on the performance of the SSCPV/T-heat exchanger system. It can be seen in Fig. 12(a) that the rise in the δ_{air} is correlated with an unfavorable increase in the panel temperature (from 302.84 K to 305.51 K), and a favorable increase in the $T_{cf,he}$ (from 301.37 K to 301.96 K). Raising the δ_{air} increases the conductive thermal resistance between the glass 3 layer and the PV panel. This leads to a reduction in conductive heat transfer and increases the temperature difference between glass 3 and the PV panel. This increase, on the other hand, will increase the contribution of radiative heat transfer and boost the T_{pv}. Moreover, the increase in temperature of the upper layers, including the NF layer, enhances the heat transfer rate in the heat exchanger, resulting in a slight increase in $T_{cf,he}$. Fig. 12(b) indicates that the increase in the δ_{air} is related to an increase in the η_{th} (from 62.09 % to 73.47 %), and η_{ov} (from 90.62 % to 100.94 %), and a decrease in the η_{el} (from 10.84 % to 10.44 %). The increase in the $T_{cf,he}$ and the increase in the T_{pv} can be introduced as the reason for the observed pattern of the changes in the η_{th} and η_{el}. The rise of the η_{ov} with the increase in δ_{air} is also due to the more significant effect of the η_{th} on the η_{ov} compared to the η_{el}. Fig. 12(c) also

shows the upward trend of σ_{th} (from 0.85 % to 1.17 %), the downward trend of σ_{el} (from 11.61 % to 11.18 %), and the upward-downward trend of σ_{ov} with increasing δ_{air}. The highest σ_{ov} equals 12.68 % which occurs at $\delta_{air} = 10$ mm.

6. Conclusions and perspectives

The present numerical research is devoted to the annual performance evaluation of an SSCPV/T-heat exchanger system that is responsible for supplying electricity and hot water to a building. The spectral splitter used in this system is a combination of NF and nanoPCM. Water with a temperature of 25 °C enters the SSCPV/T system and is heated by receiving heat from the PV panel. The hot water coming out of the SSCPV/T system enters the heat exchanger and is reheated by taking heat from the NF coming out of the SSCPV/T system. The performance of the system was investigated from the viewpoints of energy and exergy, and the effect of various factors on the annual average T_{pv}, the temperature of the water leaving the system, and the thermal, electrical, and overall efficiencies of the system are investigated. The most important results obtained from these surveys are presented below:

- The rise of L entails an increase in the T_{pv}, $T_{cf,he}$, η_{el}, σ_{th}, σ_{el} and σ_{ov}, and a decrease in η_{th} and η_{ov}.
- As the UA increases, the T_{pv} remains constant, $T_{cf,he}$, η_{th}, η_{ov} and σ_{th} increase, while η_{el}, σ_{el} and σ_{ov} decrease slightly.
- With the increase in $\dot{m}_{of}$, the $T_{cf,he}$, η_{th} and σ_{th} increase, T_{pv}, η_{el} and σ_{el} decrease, while the change trend of η_{ov} and σ_{ov} is increasing-decreasing.
- When the $\dot{m}_{cf}$ increases, the T_{pv}, $T_{cf,he}$, η_{el}, σ_{th}, σ_{ov} decrease, while the η_{th} and η_{ov} increase.
- The increase in δ_{of} is accompanied by a decrease in T_{pv}, η_{el}, η_{ov}, σ_{el}, σ_{ov} and an increase in $T_{cf,he}$ and η_{th}, σ_{th}.
- The elevation of δ_{pcm} causes a decrement in the T_{pv}, η_{el}, η_{ov}, σ_{el}, σ_{ov}, while the variation trend of $T_{cf,he}$ and η_{th} and σ_{th} is decending-asecnding.
- Boosting the δ_{cf} results in the elevation of T_{pv}, $T_{cf,he}$, η_{th}, η_{ov} and σ_{th} and decrease of η_{el}, σ_{el}, while the changing pattern of σ_{ov} is increasing-decreasing.
- As the δ_{air} augments, the T_{pv} , $T_{cf,he}$, η_{th}, η_{ov}, and σ_{th} increase, while the η_{el} and σ_{el} decrease. Also, σ_{ov} follows the upward-downward pattern with the rise of δ_{air}.
- Among the examined cases, the highest η_{ov} equals 123.87 %, which belongs to the case of $L = 1$ m, $\dot{m}_{of} = \dot{m}_{cf} = 0.05$ kg/s, $UA = 125$ W/K, $\delta_{of} = \delta_{cf} = \delta_{abs} = 10$ mm and $\delta_{pcm} = 5$ mm, while the highest σ_{ov} equals 17.08 %, which belongs to the case of $L = 1$ m, $\dot{m}_{of} = 0.05$ kg/s, $\dot{m}_{cf} = 0.005$ kg/s, $UA = 125$ W/K, $\delta_{of} = \delta_{cf} = \delta_{pcm} = \delta_{abs} = 10$ mm.

Looking forward, several research directions emerge from this study. Optimization of system parameters, such as L, UA, and the mass flow rates $\dot{m}_{of}$ and $\dot{m}_{cf}$, could lead to further improvements in efficiency. Furthermore, exploring new materials for the spectral splitter, NF, and NEPCM could result in better thermal and electrical performance, and assessing the long-term stability and durability of these materials would be valuable. Additionally, investigating the integration of the SSCPV/T system with other renewable energy sources, like wind or geothermal energy, could provide a more comprehensive solution for building energy needs. Detailed economic analyses and environmental impact assessments are essential to understand the feasibility and sustainability of the widespread adoption of this system. Conducting extensive field tests in various climatic regions would provide valuable real-world performance data, which could inform further improvements and adaptations. Developing advanced models to simulate the system's behavior under different scenarios and predict its performance over time can provide deeper insights and aid in the design of more efficient systems. It should also be noted that several practical challenges must be addressed to ensure the long-term performance of the system. Nanofluid stability is critical, as nanoparticle aggregation can degrade both thermal properties and flow dynamics. Strategies such as ultrasonic vibration, the use of surfactants, and pH adjustments have been proposed to enhance stability [51]. Moreover, by enhancing nanomaterial to PCMs (NEPCMs), we can mitigate issues like supercooling and phase separation while improving thermal conductivity and cycle stability [52]. By addressing these research directions, the potential of the SSCPV/T-heat exchanger system can be further unlocked, paving the way for its practical and widespread application in supplying sustainable energy and hot water to buildings.

CRediT authorship contribution statement

Jaber Aboueian: Software, Methodology, Investigation, Data curation. **Amin Shahsavar:** Writing – original draft, Supervision, Formal analysis, Conceptualization. **Hamid Reza Askarifard Jahromi:** Software, Methodology, Investigation, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

[1] https://www.iea.org/energy-system/renewables/solar-pv.
[2] A. Kouravand, A. Kasaeian, F. Pourfayaz, M.A. Vaziri Rad, Evaluation of a nanofluid-based concentrating photovoltaic thermal system integrated with finned PCM heatsink: an experimental study, Renew. Energy 201 (2022) 1010–1025.
[3] A.K. Tiwari, K. Chatterjee, S. Agrawal, G.K. Singh, A comprehensive review of photovoltaic-thermal (PVT) technology: performance evaluation and contemporary development, Energy Rep. 10 (2023) 2655–2679.
[4] X.Z. Wu, High-efficiency polycrystalline CdTe thin-film solar cells, Sol. Energy 77 (2004) 803–814.
[5] E.D. Kosten, J.H. Atwater, J. Parsons, A. Polman, H.A. Atwater, Highly efficient GaAs solar cells by limiting light emission angle, Light Sci. Appl. 2 (2013) e45.
[6] M. Kaelin, D. Rudmann, A.N. Tiwari, Low cost processing of CIGS thin film solar cells, Sol. Energy 77 (2004) 749–756.
[7] G.R. Na, L.J. Zhang, Stable and luminescent halide perovskite fabricated in water, Light Sci. Appl. 9 (2020) 106.
[8] F. Wang, Z. Li, M. Liu, X. Liu, D. Pang, W. Du, X. Cheng, Y. Zhang, W. Guo, Heat-dissipation performance of photovoltaic panels with a phase-change-material fin structure, J. Clean. Prod. 423 (2023) 138756.
[9] E. Jamali, M.H. Nobakhti, B.M. Ziapour, M. Khayat, Performance analysis of a novel model of photovoltaic PV-TEGs system enhanced with flat plate mirror reflectors, Energy Convers. Manag. 279 (2023) 116766.
[10] M. Jafaryar, M. Sheikholeslami, Efficacy of magnetic field on performance of photovoltaic solar system utilizing ferrofluid, J. Magn. Magn Mater. 562 (2022) 169798.
[11] A. Lashin, Analysis of thermal performance a heat pipe for concentrated photovoltaic cooling, Journal of Radiation Research and Applied Sciences 16 (2023) 100606.
[12] P. Azimi, A. Shahsavar, N. Azimi, Improving the performance of photovoltaic modules using ultrasonic- thermoelectric generators, Appl. Therm. Eng. 220 (2023) 119704.
[13] W.E. Ewe, A. Fudholi, K. Sopian, R. Moshery, N. Asim, W. Nuriana, A. Ibrahim, Thermo-electro-hydraulic analysis of jet impingement bifacial photovoltaic thermal (JIBPVT) solar air collector, Energy 254 (2022) 124366.
[14] M. Raju, R.N. Sarma, A. Suryan, P.P. Nair, S. Nizetic, Investigation of optimal water utilization for water spray cooled photovoltaic panel: a three-dimensional computational study, Sustain. Energy Technol. Assessments 51 (2022) 101975.
[15] A. Shahsavar, A. Jahangiri, A. Qatarani nejad, G. Ahmadi, A. Karamzadeh dizaji, Energy and exergy analysis and multi-objective optimization of using combined vortex tube-photovoltaic/thermal system in city gate stations, Renew. Energy 196 (2022) 1017–1028.
[16] A. Shahsavar, A.H.A. Alwaeli, N. Azimi, S. Rostami, K. Sopian, M. Arici, P. Estelle, S. Nizetic, A. Kasaeian, H.M. Ali, Z. Ma, M. Afrand, Exergy studies in water-based and nanofluid-based photovoltaic/thermal collectors: Status and prospects, Renew. Sustain. Energy Rev. 168 (2022) 112740.
[17] H. Duwairi, M. Qasem, Corrugating photovoltaic modules enhances thermal efficiency and power output, Sol. Energy 188 (2019) 318–326.
[18] A. Kazemian, M. Khatibi, S. Entezari, T. Ma, H. Yang, Efficient energy generation and thermal storage in a photovoltaic thermal system partially covered by solar cells and integrated with organic phase change materials, Renew. Sustain. Energy Rev. 188 (2023) 113705.
[19] C. Sirin, J. Gogginns, M. Hajdukiewicz, A review on building-integrated photovoltaic/thermal systems for green buildings, Appl. Therm. Eng. 229 (2023) 120607.
[20] l. Gao, X. Zhang, W. Hua, Recent progress in photovoltaic thermal phase change material technology: a review, J. Energy Storage 65 (2023) 107317.
[21] J. oh, S. Bae, H. Chae, J. Jeong, Y. Nam, Photovoltaic–thermal advanced technology for real applications: review and case study, Energy Rep. 10 (2023) 1409–1433.
[22] A. Ustaoglu, V. Akgul, J. Okajima, Performance investigation of truncated low concentrating photovoltaic-thermal systems with V-trough, compound hyperbolic and compound parabolic concentrators, Appl. Therm. Eng. 232 (20230 121028.
[23] M.A. Fikri, M. Samykano, A.K. Pandey, K. Kadirgama, R.R. Kumar, J. Selvaraj, N.A. Rahim, V.V. Tyagi, K. Sharma, R. Saidur, Recent progresses and challenges in cooling techniques of concentrated photovoltaic thermal system: a review with special treatment on phase change materials (PCMs) based cooling, Sol. Energy Mater. Sol. Cell. 241 (2022) 111739.
[24] G. Huang, K. Wang, C.N. Markides, Efficiency limits of concentrating spectral-splitting hybrid photovoltaic-thermal (PV-T) solar collectors and systems, Light Sci. Appl. 10 (2021) 28.
[25] W. Hong, B. Li, X. Niu, Y. Li, J. Lan, Recent progress in thermal energy recovery from the decoupled photovoltaic/thermal system equipped with spectral splitters, Renew. Sustain. Energy Rev. 167 (2022) 112824.
[26] F. Yazdanifard, M. Ameri, R. Taylor, Parametric investigation of a nanofluid-NEPCM based spectral splitting photovoltaic/thermal system, Energy Convers. Manag. 240 (2021) 114232.
[27] W.C. Meraje, C.C. Huang, J. Barman, C.Y. Huang, C.F.J. Kuo, Design and experimental study of a Fresnel lens-based concentrated photovoltaic thermal system integrated with nanofluid spectral splitter, Energy Convers. Manag. 258 (2022) 115455.
[28] j. Huang, X. Han, X. Zhao, A.A. Khosa, C. Meng, The stability, optical behavior optimization of $Ag@SiO_2$ nanofluids and their application in spectral splitting photovoltaic/thermal receivers, Renew. Energy 190 (2022) 865–878.
[29] O. Elharoun, M. Tawfik, I.I. El-Sharkawy, E. Zeidan, Experimental investigation of photovoltaic performance with compound parabolic solar concentrator and fluid spectral filter, Energy 278 (2023) 127848.
[30] L. Zhang, Y. Hu, J. Chu, C. Xu, Y. Yao, H. Lv, Performance investigation on a concentrating photovoltaic thermal system integrated with spectral splitter and absorption heat pump, Appl. Therm. Eng. 237 (2024) 121772.
[31] M. Farsijani, H. Akbari, M. Ghasemi, Enhanced performance and stability of a solar pond using an external heat exchanger filled with nano-phase change material, J. Build. Eng. 55 (2024) 104–115.
[32] S. Hassani, R. Saidur, S. Mekhilef, R.A. Taylor, Environmental and exergy benefit of nanofluid-based hybrid PV/T systems, Energy Convers. Manag. 123 (2016) 431–444.
[33] T.P. Otanicar, R.A. Taylor, C. Telang, Photovoltaic/thermal system performance utilizing thin film and nanoparticle dispersion based optical filters, J. Renew. Sustain. Energy 5 (2013) 033124.
[34] F. Yazdanifard, M. Ameri, R.A. Taylor, Numerical modeling of a concentrated photovoltaic/thermal system which utilizes a PCM and nanofluid spectral splitting, Energy Convers. Manag. 215 (2020) 112927.
[35] https://www.irimo.ir/index.php?newlang=eng.
[36] F. Goia, M. Perino, M. Haase, A numerical model to evaluate the thermal behaviour of PCM glazing system configurations, Energy Build. 54 (2012) 141–153.
[37] K.A.R. Ismail, C.T. Salinas, J.R. Henriquez, Comparison between PCM filled glass windows and absorbing gas filled windows, Energy Build. 40 (2008) 710–719.
[38] W.M. Kays, A.L. London, Compact Heat Exchangers, McGraw Hill, New York, USA, 1984.
[39] F.P. Fleming, L.D.A. Silva, G.S.V. Lima, I. Herzog, H.R.B. Orlande, J.-L. Daridon, J. Pauly, L.F.A. Azevedo, Thermal conductivity of heavy, even-carbon number n-alkanes (C22 to C32), Fluid Phase Equilib. 477 (2018) 78–86.
[40] S. Himran, A. Suwono, G.A. Mansoori, Characterization of alkanes and paraffin waxes for application as phase change energy storage medium, Energy Sources 16 (1994) 117–128.
[41] S.H. Guang, Q. Hai-Yan, Z. Qingdong, Rayleigh, Mie, and Tyndall scatterings of polystyrene microspheres in water: wavelength, size, and angle dependences, J. Appl. Phys. 105 (2009) 023110.
[42] C.F. Bohren, D.R. Huffman, Absorption and Scattering of Light by Small Particles: John Wiley & Sons, 2008.
[43] R.A. Taylor, T.P. Otanicar, Y. Herukerrupu, F. Bremond, G. Rosengarten, E.R. Hawkes, X. jiang, S. Coulombe, Feasibility of nanofluid-based optical filters, Appl. Opt. 52 (2013) 1413–1422.
[44] J.R. Howell, M.P. Menguc, R. Siegel, Thermal Radiation Heat Transfer, CRC Press, 2015.
[45] M.F. Modest, Radiative Heat Transfer, Academic Press, 2013.

[46] H. Weinlader, Optische Charakterisierung von Latentwarmespeichermaterialien, zur, Tageslichtnutzung (2003).
[47] A.K. Gonzalez-Alcalde, E.R. Mendez, E. Teran, F.L.S. Cuppo, J.A. Olivares, A. Garcia-Valenzuela, Reflection of diffuse light from dielectric one-dimensional rough surfaces, Journal of Optical Society of America A 33 (2016) 373–382.
[48] M. Muhieddine, E. Canot, R. March, Various Approaches for Solving Problems in Heat Conduction with Phase Change, 2009.
[49] Y. Abbassi, E. Baniasadi, H. Ahmadikia, Transient energy storage in phase change materials, development and simulation of a new TRNSYS component, J. Build. Eng. 50 (2022) 104188.
[50] C. Appendix, Physical properties of liquids and Gases, in: A.K. Coker (Ed.), Ludwig's Applied Process Design for Chemical and Petrochemical Plants, fourth ed., Gulf Professional Publishing, Burlington, 2007, pp. 827–862.
[51] L. Sun, L. Yang, N. Zhao, J. Song, X. Li, X. Wu, A review of multifunctional applications of nanofluids in solar energy, Powder Technol. 411 (2022) 117932.
[52] Q. Zhang, F. Ma, L. Liu, W. Tan, M. Jing, L. Wang, M. Cai, H. Wang, Recent advances in nano-enhanced phase change materials, J. Mater. Sci. 59 (2024) 5247–5267.